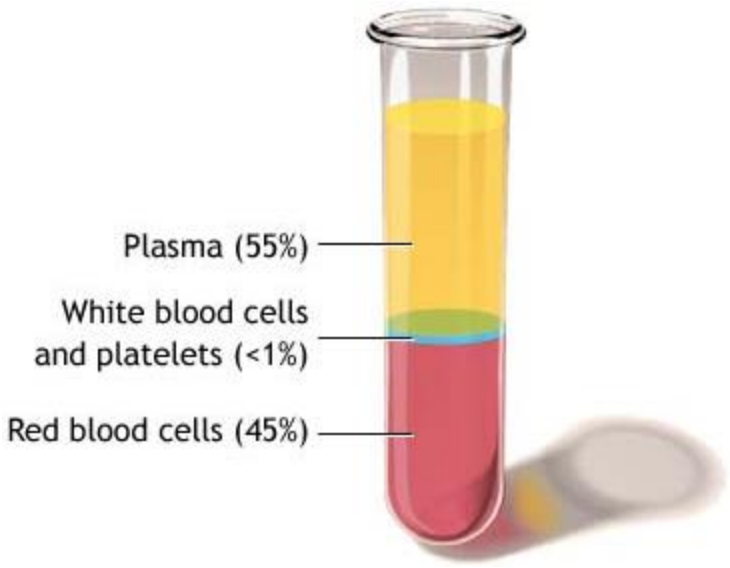
Simulations of electrodeformation in blood analyzers:

Maxwell Stress in an IBM solver

D. Di Pietro, S. Mendez, **A. Spadotto**

IMAG, CNRS, Montpellier

Blood Analysis

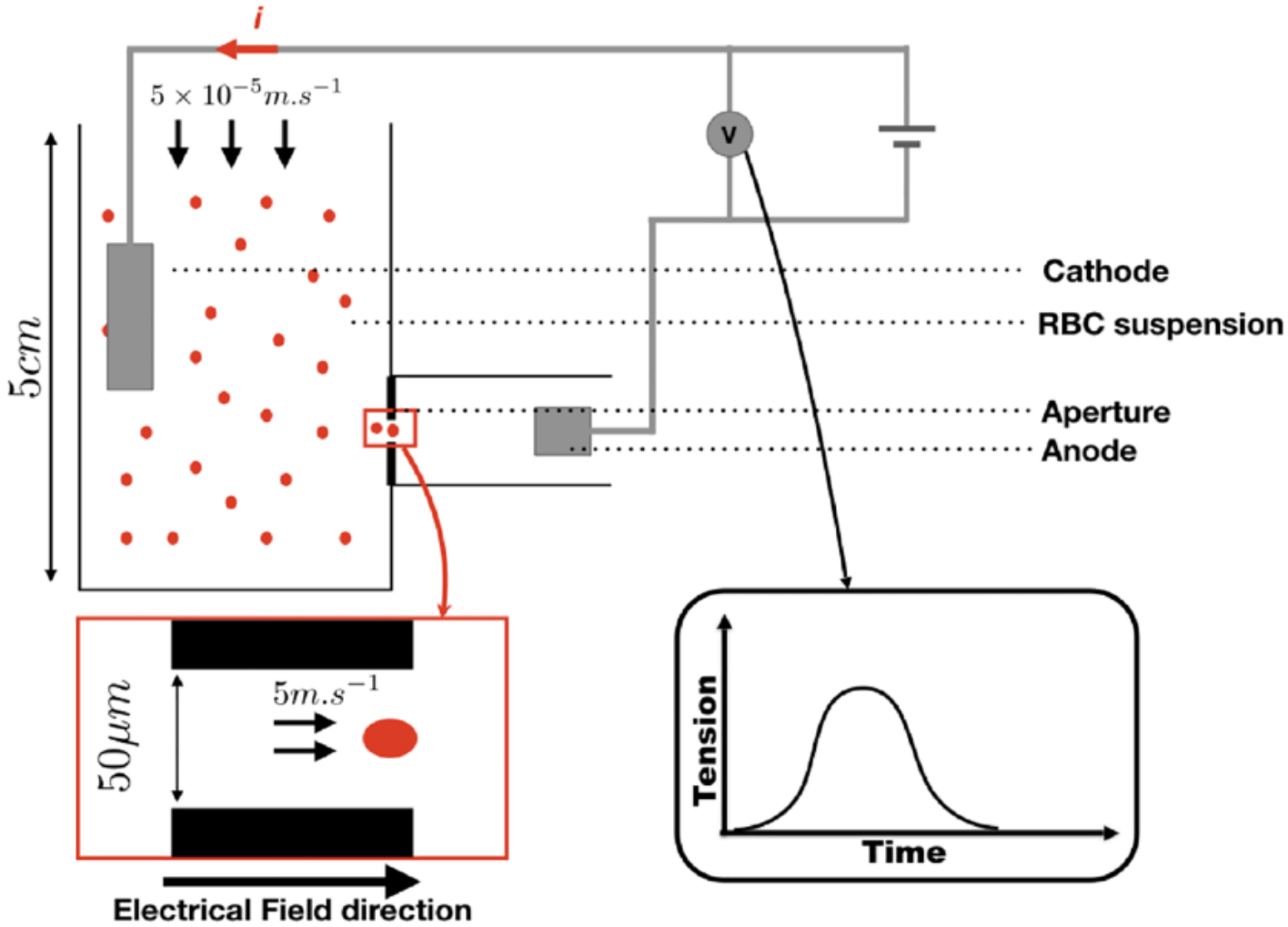


Complete blood count	Result	Reference values
WBC	7.36 × 10.e9/L	(4-11)
RBC	5.12 ×10.e12	(4.2-5.5)
HGB	15.4 g/dl	(12-16)
HCT	45%	(37-47)
MCV	87.9 fl	(80-94)
MCH	30 pg	(27-32)
MCHC	34 g/dl	(32-36)
RDW	11.4 %	(11.5-14.5)
Platelet count	183 × 10.e9/L	(140-450)
MPV	9.43 fl	(7.2-11.1)

RBC: Red Blood Count
MCV: Mean Cell Volume

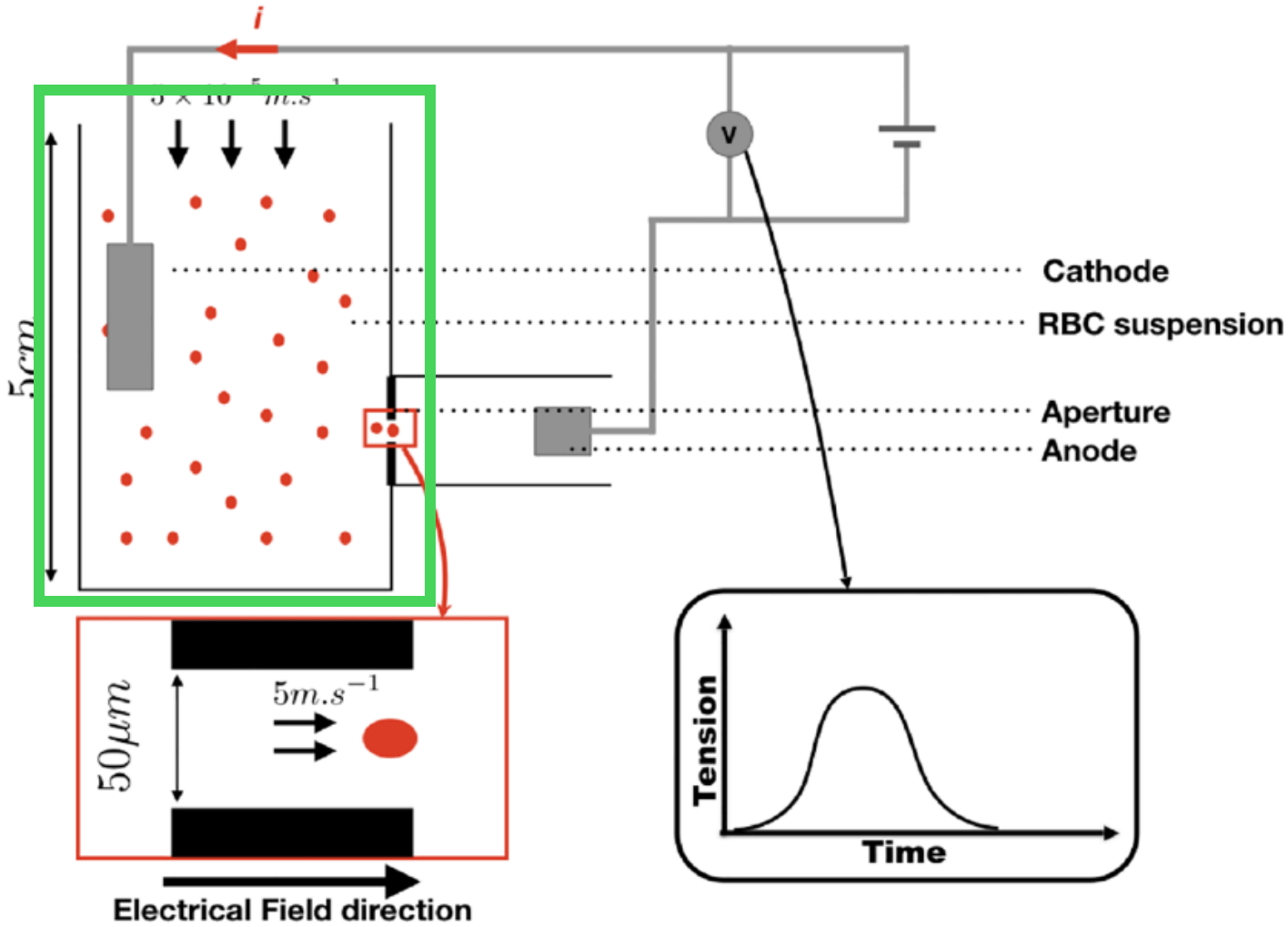
RBC and MCV: indicator of **anemia**

Coulter Counters

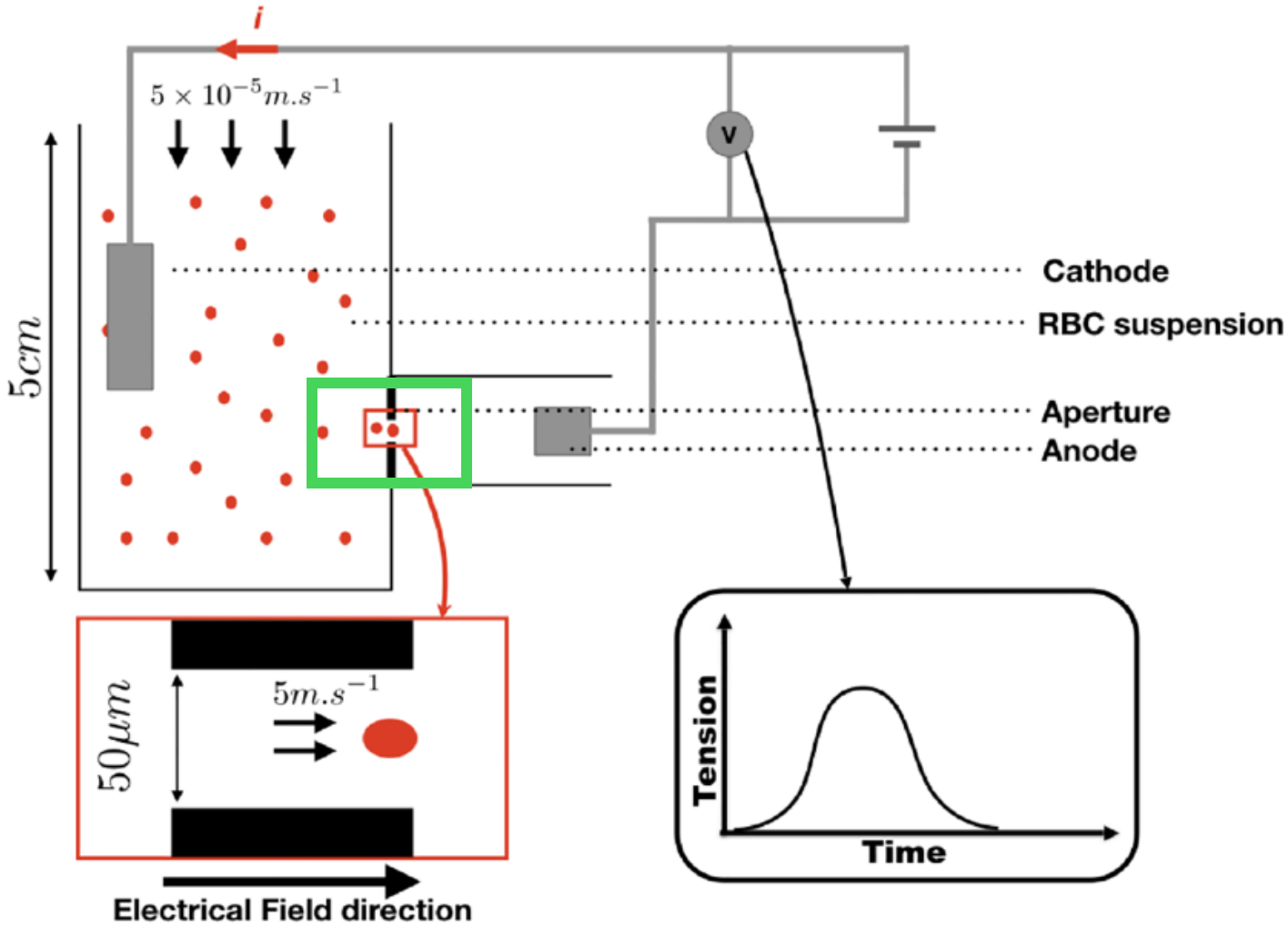


W. Coulter, patent in 1953

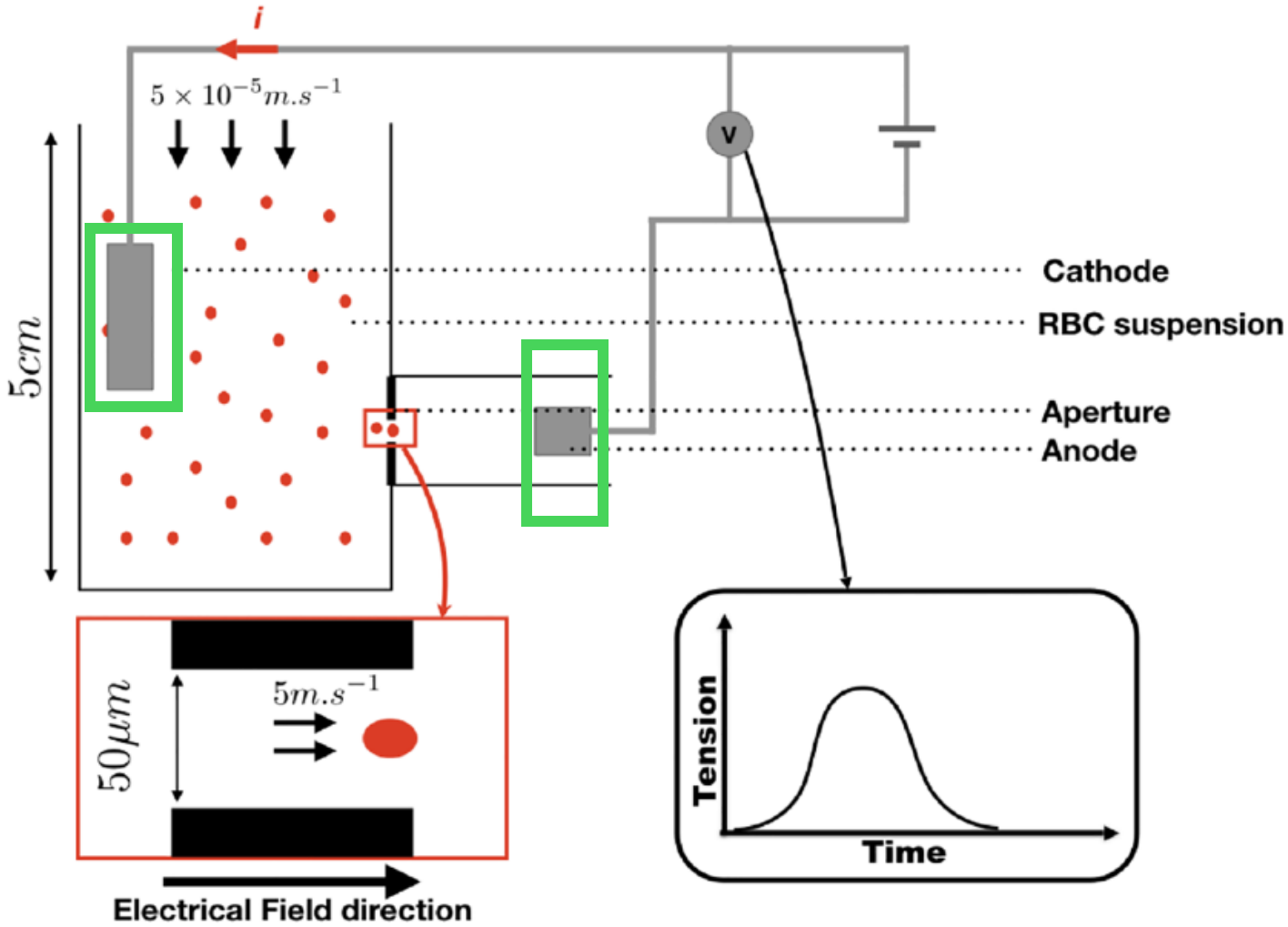
Coulter Principle



Coulter Principle



Coulter Principle

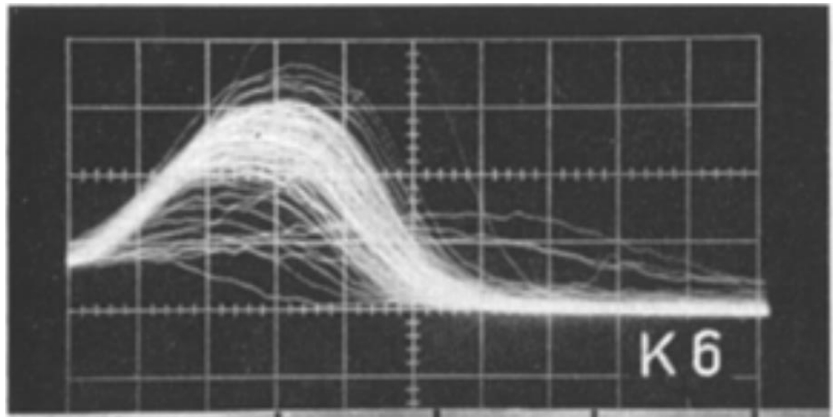


History of a Signal

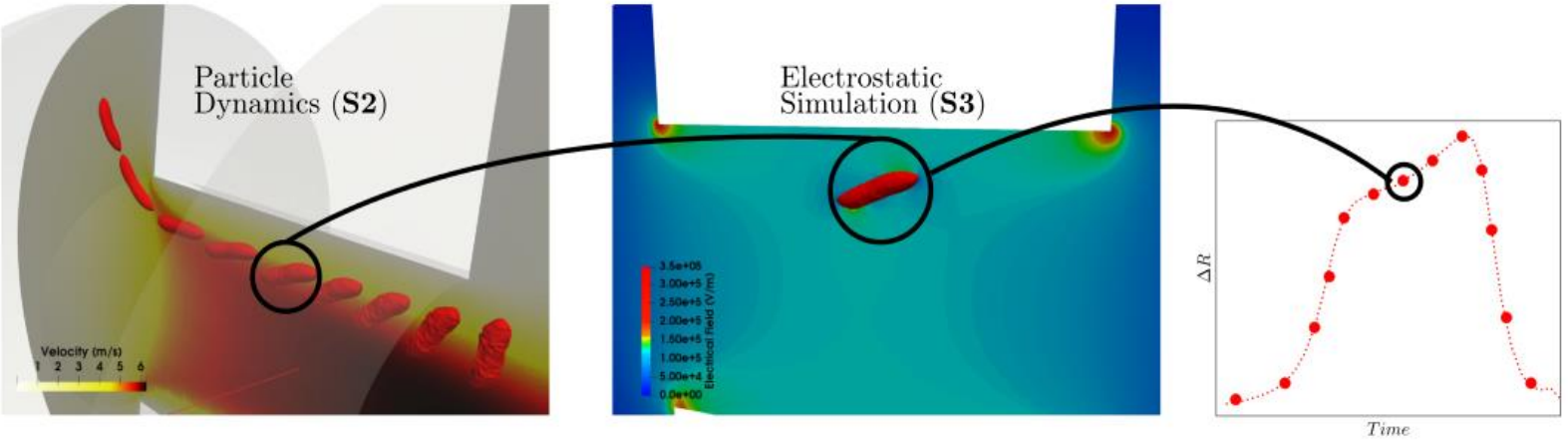
Kachel, 1970
Volume Estimation

$$\Delta\Phi = \boxed{k_s} E^2 V_{RBC}$$

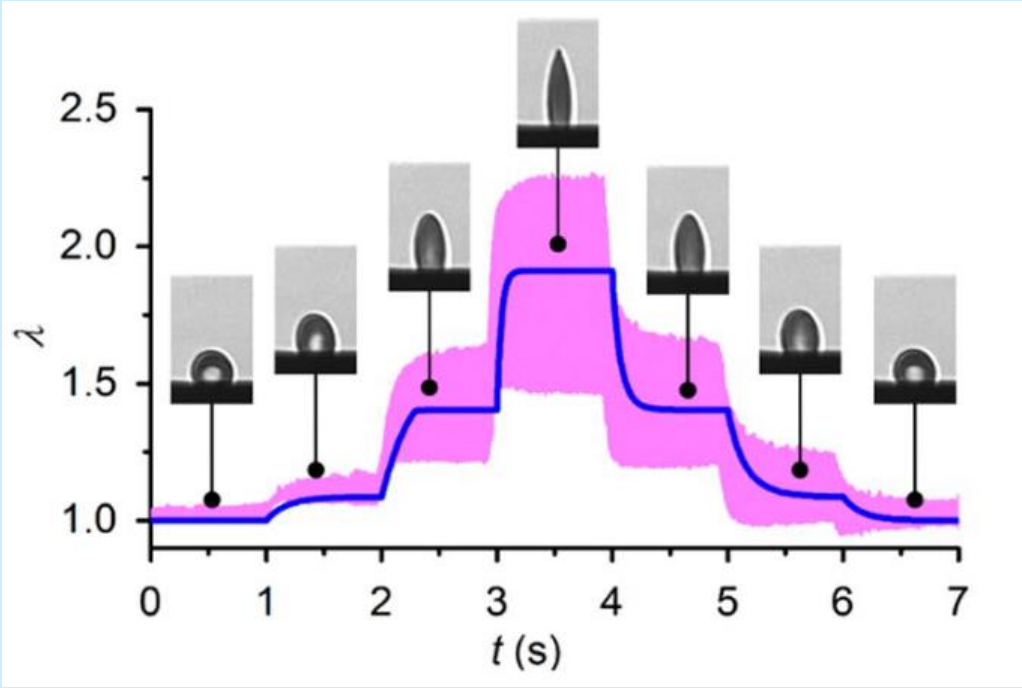
Sensibility to
Trajectory Perturbation and
Orientation



Taraconat et al, 2020
Numerical Simulation
&
Trajectory Detection

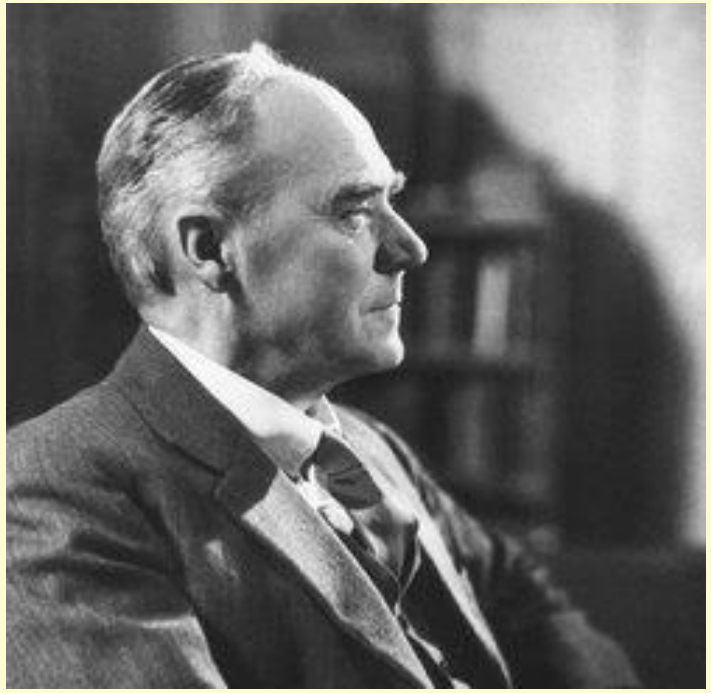


Electrodeformation



Modeling erythrocyte electrodeformation in response to amplitude modulated electricwaveforms

Yuhao Qiang, Jia Liu, Fan Yang, Darryl Dieujuste & E. Du



"In addition to the mechanical pressure and viscous stress, there is an electrical force due to the free-charge density and due to polarization"

G.I. Taylor, ELECTROHYDRODYNAMICS: A REVIEW OF THE ROLE OF INTERFACIAL SHEAR STRESSES, 1966

Program

1) Immersed Boundary Simulations: standard for Membrane hydrodynamics

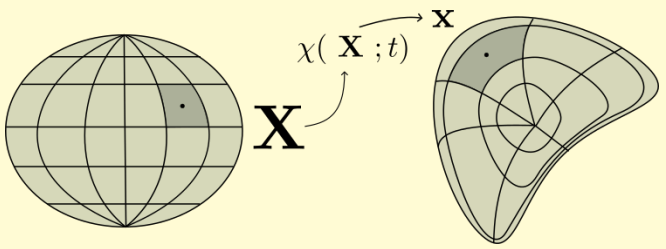
2) PDE model of Maxwell Stress

3) Strategies to merge Maxwell stress and IBM

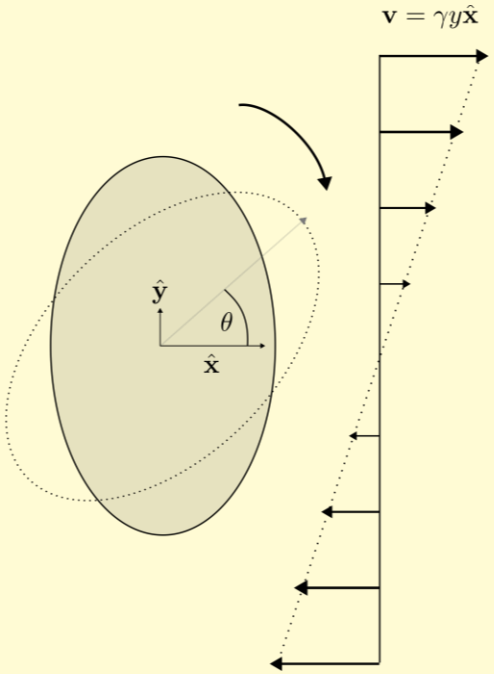
4) Results on Benchmark Configurations

RBC physics

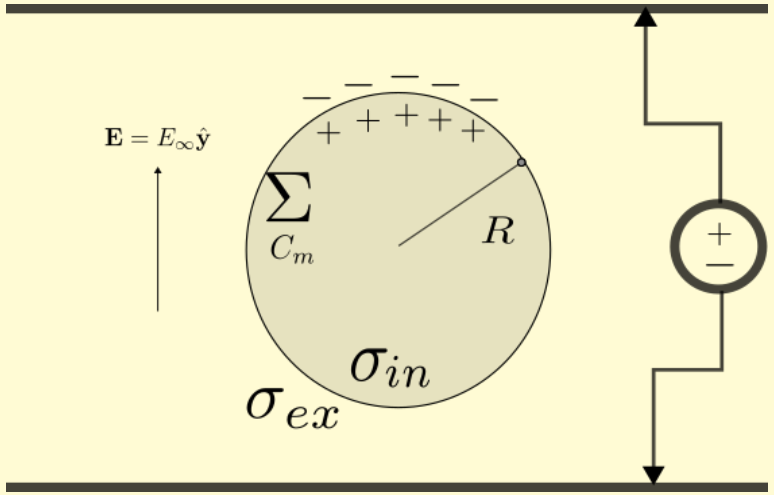
Membrane Model



Fluid-Structure Interaction

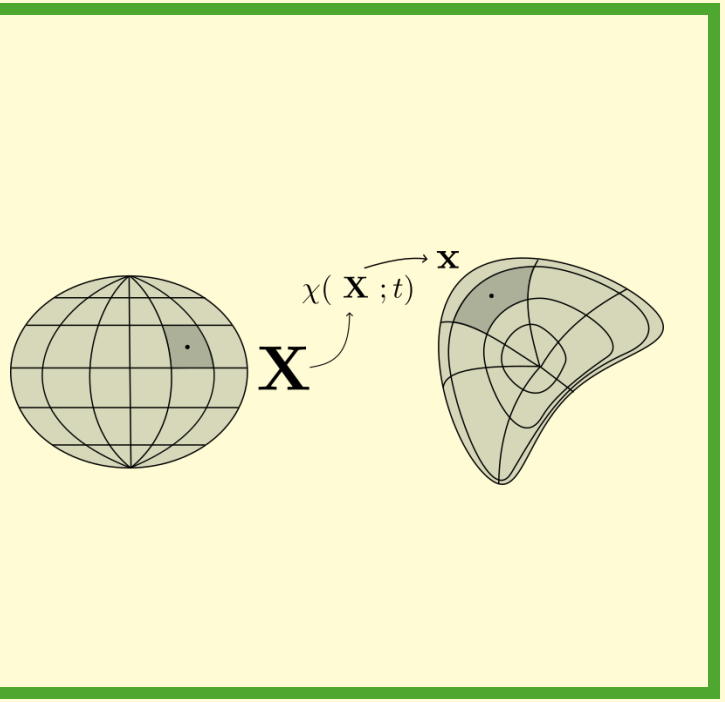


Electrohydrodynamics

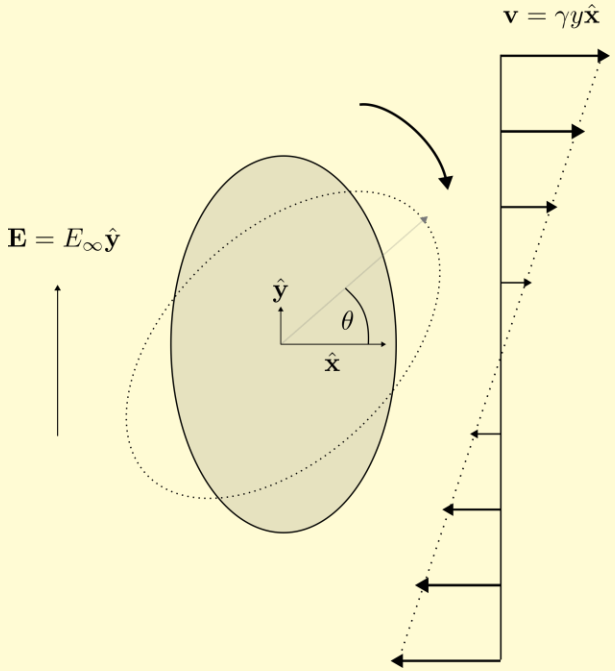


RBC physics

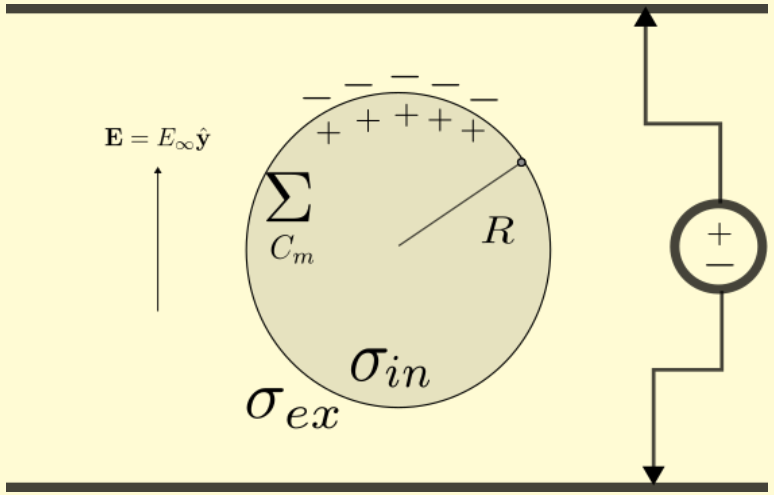
Membrane Model



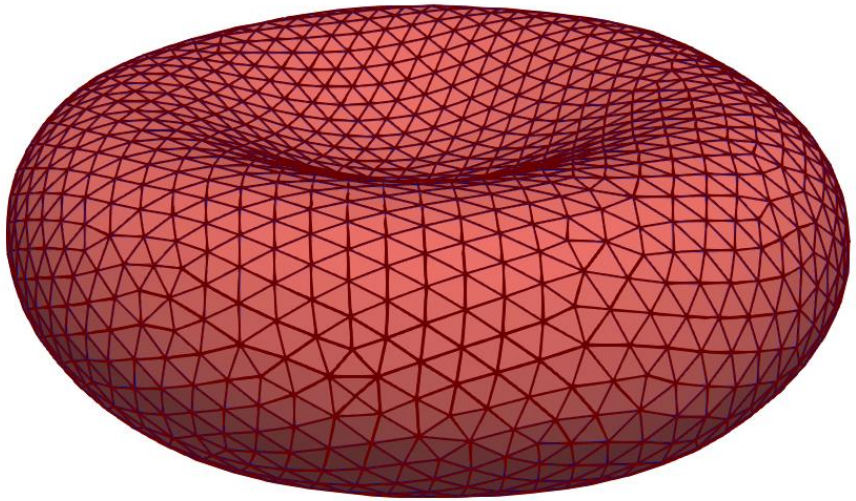
Fluid-Structure Interaction



Electrohydrodynamics



Membrane Discretization



A mesh for the Surface

Skalak Energy+Helfrich Energy: dependency on configuration

$$W_{SK} = \frac{G_{SK}}{4}[(\lambda_1^2 + \lambda_2^2 - 2)^2 + 2(\lambda_1^2 + \lambda_2^2 - \lambda_1\lambda_2 - 1) + C(\lambda_1^2\lambda_2^2 - 1)^2]$$

$$W_{hel} = \frac{k_b}{2}\int_\Sigma (H - C_0)^2 d\Sigma$$

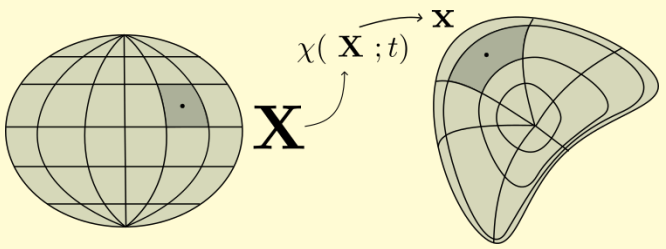
Configuration-dependent Forces living on nodes; velocities

$$\boldsymbol{X}_i, \boldsymbol{V}_i, \boldsymbol{F}_i$$

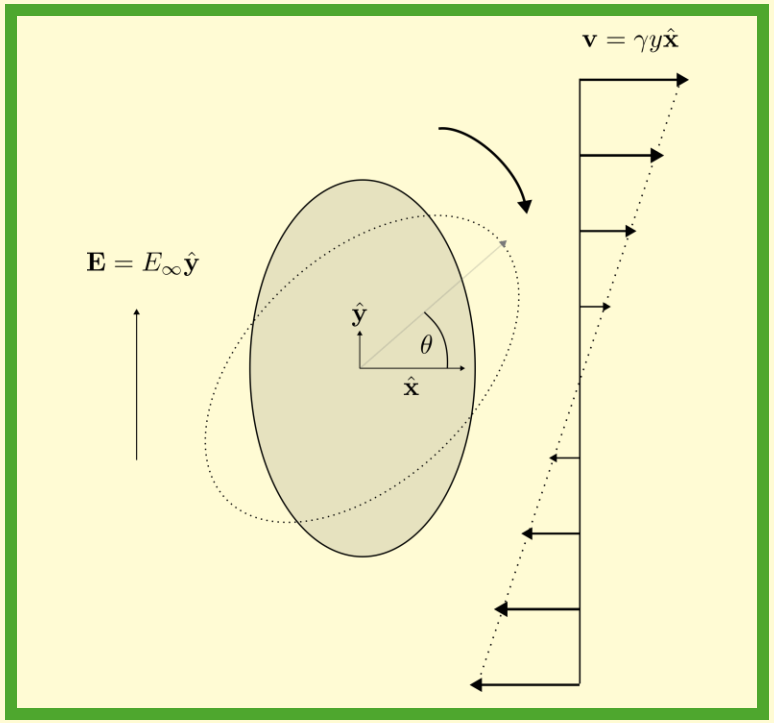
$$\dot{\boldsymbol{X}}_i = \boldsymbol{V}_i$$

RBC physics

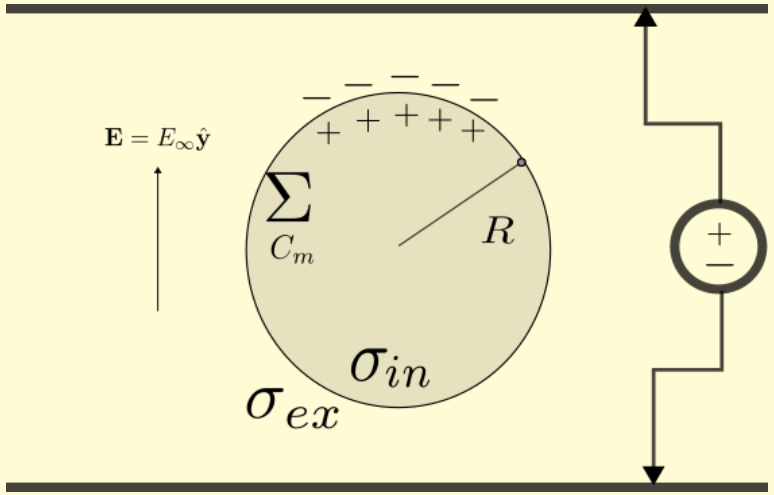
Membrane Model



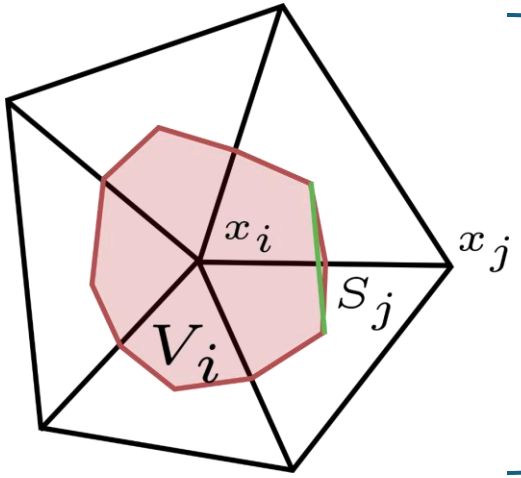
Fluid-Structure Interaction



Electrohydrodynamics

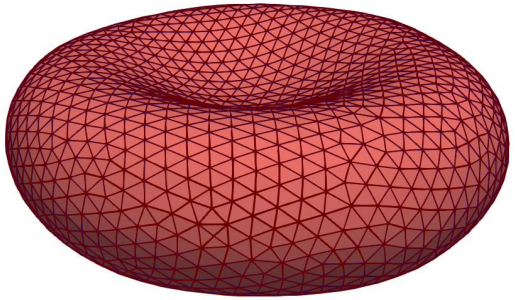


Immersed Boundary Method



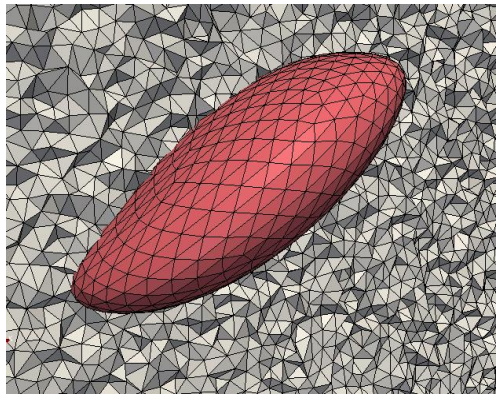
Finite Volumes

$$\boldsymbol{x}_i, p_i, \boldsymbol{v}_i$$

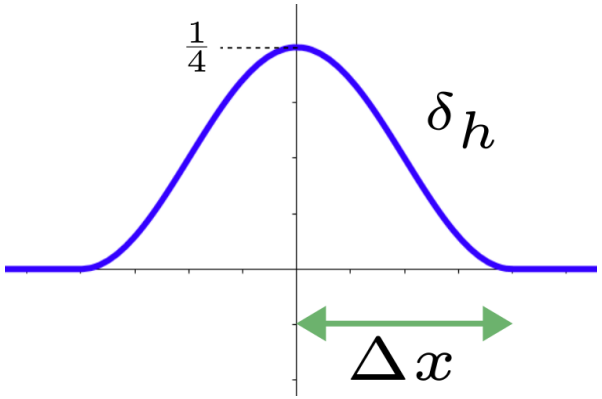


$$\boldsymbol{X}_i, \boldsymbol{V}_i, \boldsymbol{F}_i$$

$$\dot{\boldsymbol{X}}_i = \boldsymbol{V}_i$$



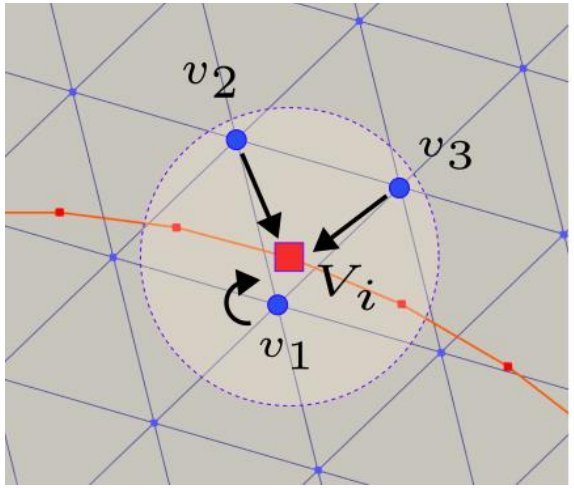
Velocity interpolation

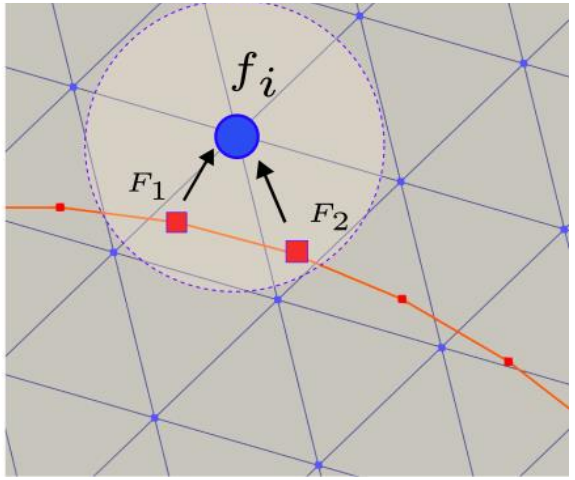


Force Transmission

$$\boldsymbol{V}_i = \sum_j \delta_h(\boldsymbol{x}_j - \boldsymbol{X}_i)\boldsymbol{v}_j$$

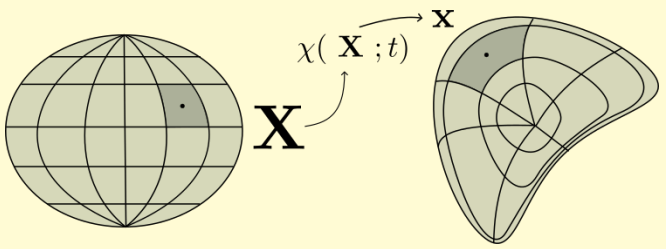
$$\boldsymbol{f}_i = \sum_j \delta_h(\boldsymbol{X}_j - \boldsymbol{x}_i)\boldsymbol{F}_j$$



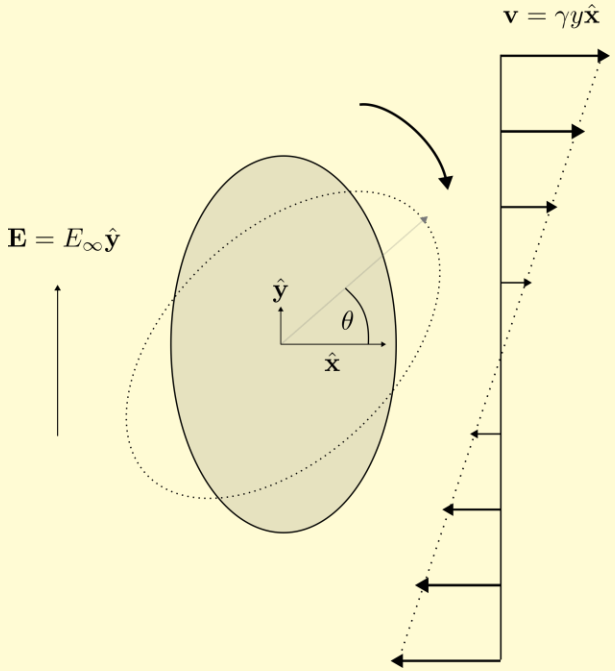


RBC physics

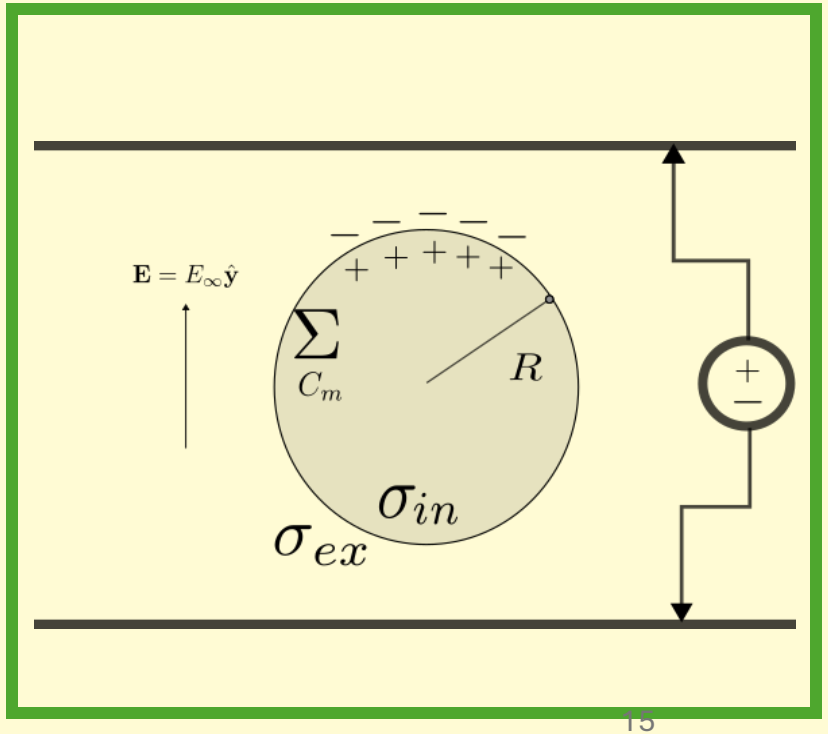
Membrane Model



Fluid-Structure Interaction



Electrohydrodynamics

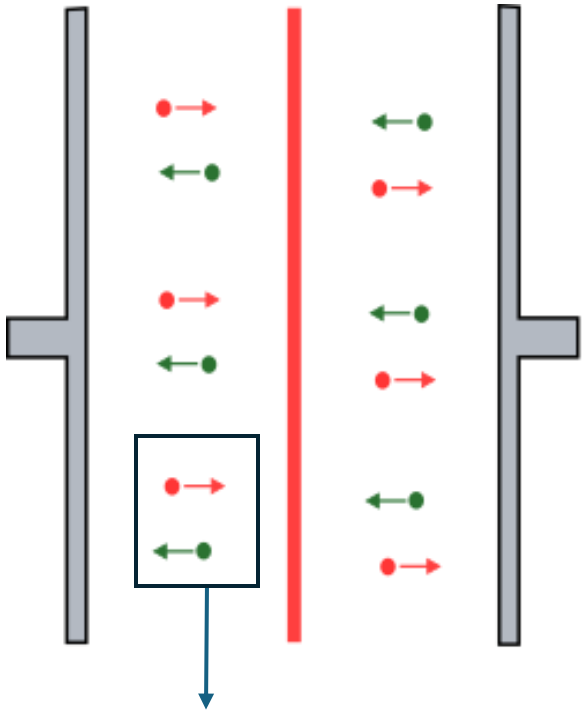


Interfacial Electric Stress

$$\rho\frac{D\boldsymbol{v}}{Dt} + \nabla\cdot(\bar{\bar{T}}) = \boldsymbol{f} \qquad \bar{\bar{T}}_{MW} = \epsilon(\boldsymbol{E}\boldsymbol{E} - \frac{1}{2}\boldsymbol{E}\cdot\boldsymbol{E}\bar{\bar{I}}) \quad \boldsymbol{f}_{MW} = [\![\bar{\bar{T}}_{MW}]\!]\cdot\boldsymbol{n}$$

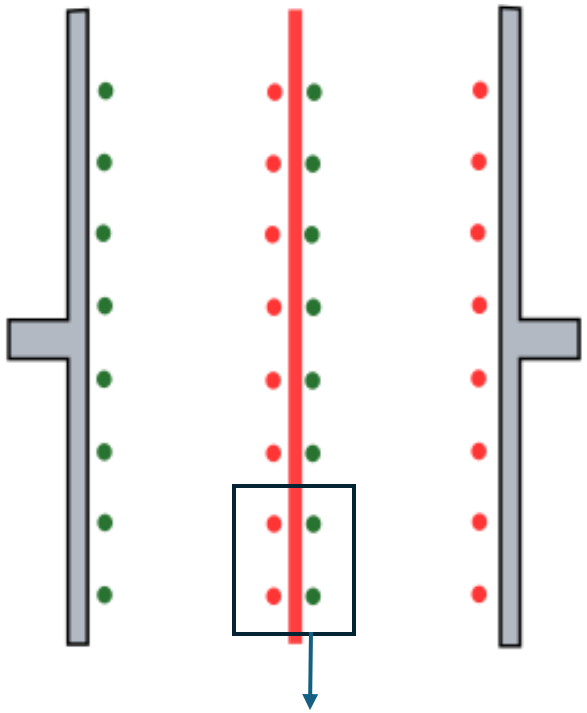
G.I. Taylor, ELECTROHYDRODYNAMICS: A REVIEW OF THE ROLE OF INTERFACIAL SHEAR STRESSES, 1966

Charge Migration



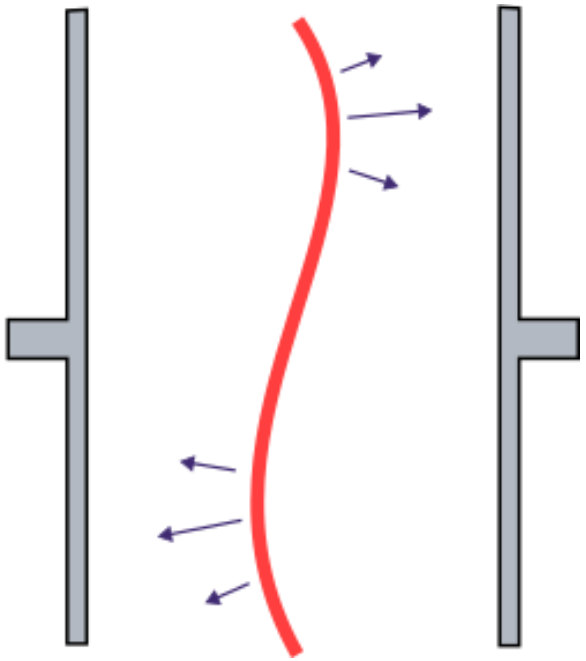
Net volume charge = 0

Interface Deposit



Net surface charge = 0

Instabilisation



Leaky Dielectric Model

Potential

$$\phi : \Omega \to \mathbb{R}$$

Electric Field

$$\boldsymbol{E} = \nabla \phi$$

Current Density (Ohm Law)

$$\boldsymbol{J} = \sigma \boldsymbol{E}$$

Galvani Potential

$$V_{\mathrm{m}} : \Sigma \to \mathbb{R}$$

$$\begin{cases} \nabla \cdot (\sigma \nabla \phi) = 0 \\ [\![\sigma \nabla \phi]\!] \cdot \boldsymbol{n} = 0 \\ [\![\phi]\!] = V_{\mathrm{m}} \\ \nabla \phi|_{\infty} = \boldsymbol{E}_{\infty} \end{cases}$$

Current continuity

Neutral Membrane

Potential Jump

Far Field

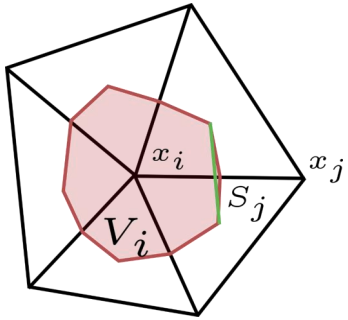
Membrane Model

$$C_m \frac{\partial V_m}{\partial t} + G_m V_m = \sigma_{in} \boldsymbol{E}_{in} \cdot \boldsymbol{n} = \sigma_{ex} \boldsymbol{E}_{ex} \cdot \boldsymbol{n}$$

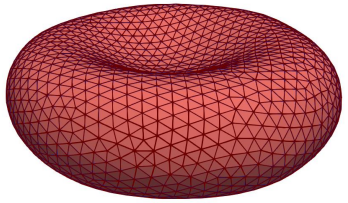
ELECTROHYDRODYNAMICS:
The Taylor-Melcher Leaky Dielectric Model
D. A. Saville, 1997

Double Mesh

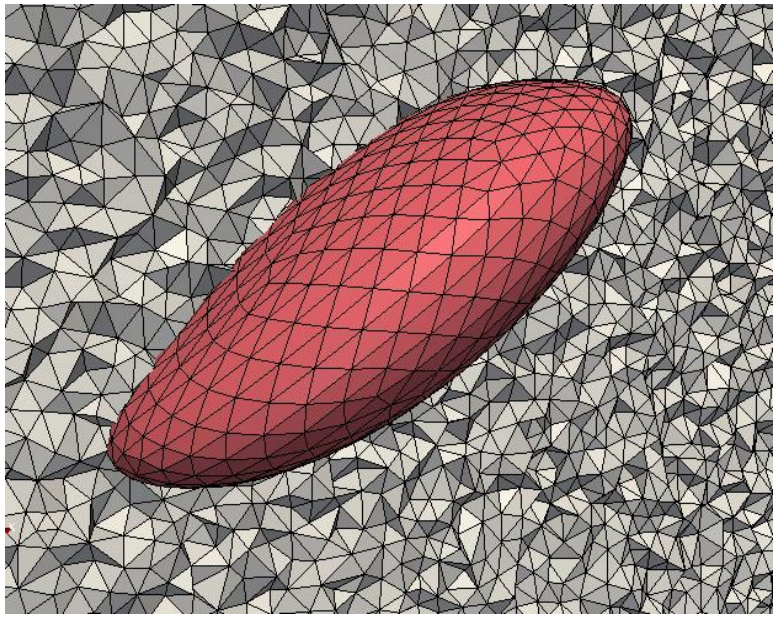
Variables on Separate Entities



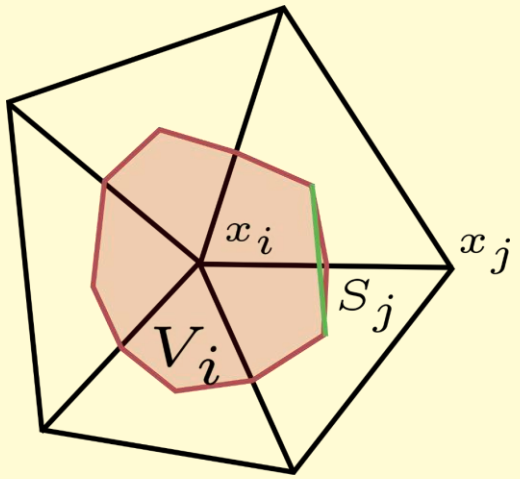
$$\boldsymbol{x}_i, \phi_i, \boldsymbol{E}_i$$



$$\boldsymbol{X}_i, V_i^{\mathrm{m}}$$



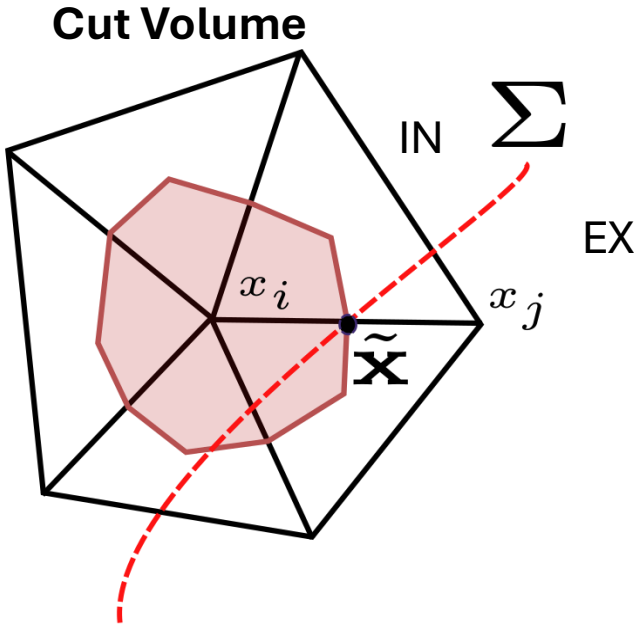
FV at Interface Volumes

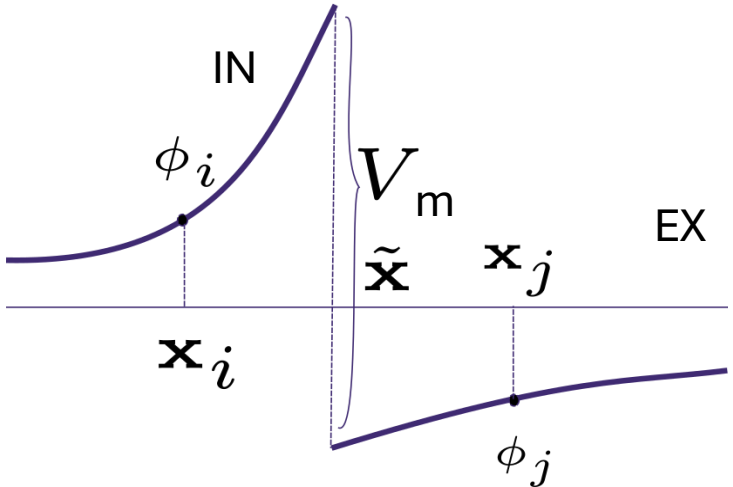


$$\nabla \cdot (\nabla \phi) = 0$$

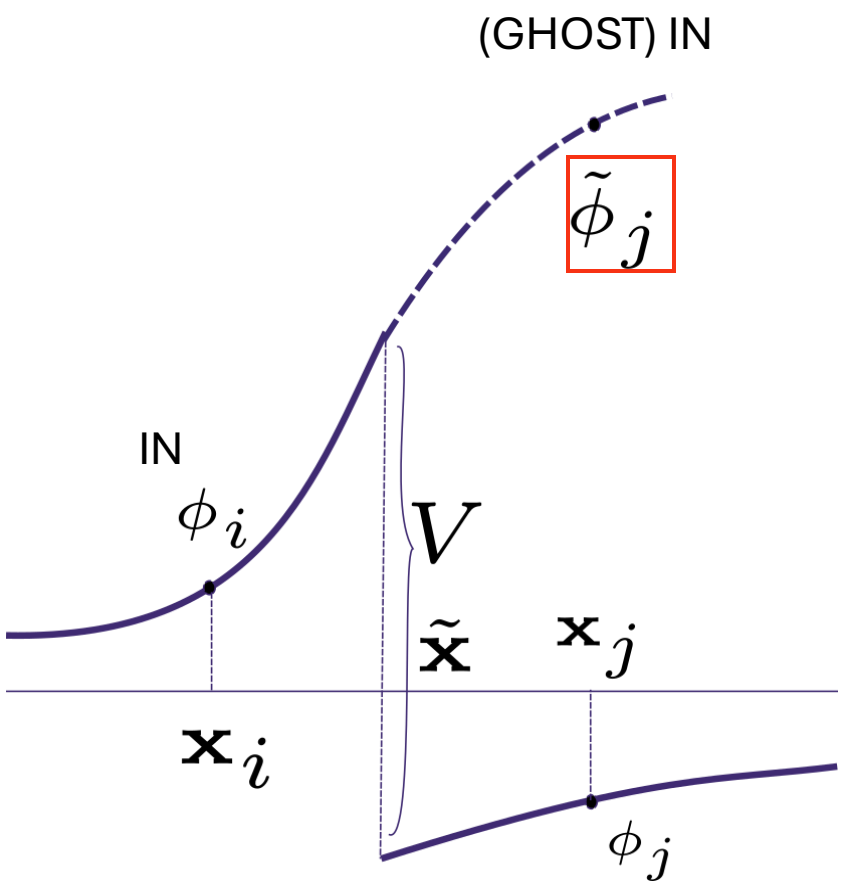
$$\sum_j \int_S \nabla \phi \cdot \boldsymbol{n} = 0$$

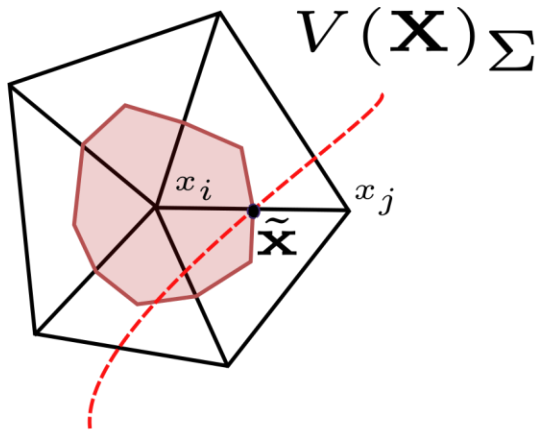
$$\sum_j (\phi_j - \phi_i) \frac{\boldsymbol{x}_j - \boldsymbol{x}_i}{\|\boldsymbol{x}_j - \boldsymbol{x}_i\|} \cdot \boldsymbol{n} |S| = 0$$

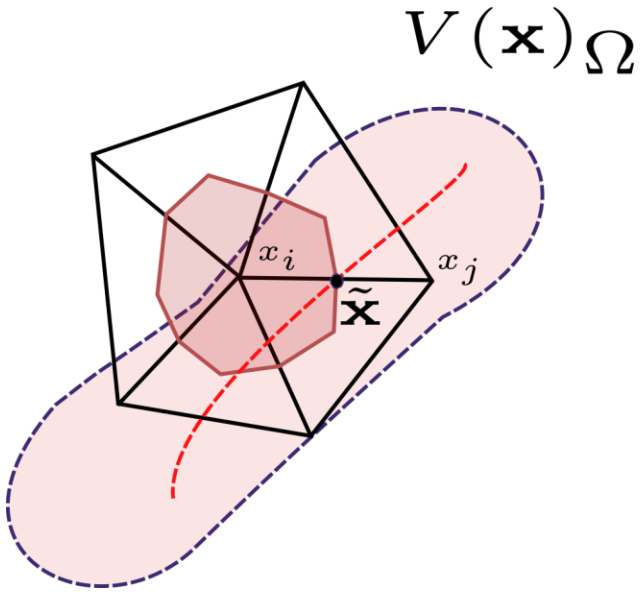




Ghost Fluid Method







$$\begin{aligned}\tilde{\phi}(\boldsymbol{x}_j) &= \phi_{in}(\boldsymbol{x}_j)\\ &= \phi_{ex}(\boldsymbol{x}_j) - (\phi_{ex}(\boldsymbol{x}_j) - \phi_{in}(\boldsymbol{x}_j))\\ &= \phi_j - V(\boldsymbol{x}_j)\end{aligned}$$

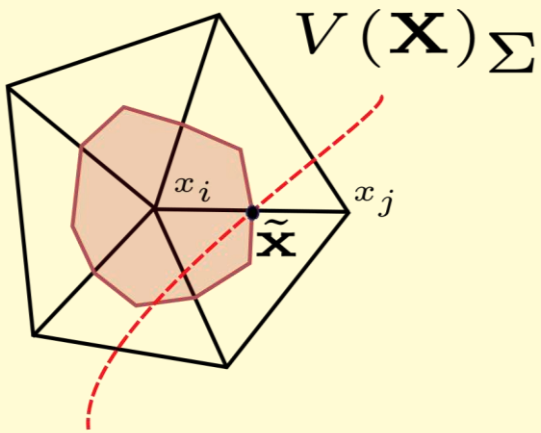
An accurate conservative level set/ghost fluid method for simulating turbulent atomization, *Desjardins, Olivier and Moureau, Vincent and Pitsch, Heinz*, 2008

Ghost Fluid Method

$$[\![\phi]\!](\boldsymbol{x}_j) = [\![\phi]\!](\tilde{\boldsymbol{x}}) + \nabla[\![\phi]\!] \cdot (\boldsymbol{x}_j - \tilde{\boldsymbol{x}})$$ **Taylor Exp**

$$= V_m(\tilde{\boldsymbol{x}}) - \frac{[\![\sigma]\!]}{\sigma_{ex}}(\phi_{,n}^{in})\|\boldsymbol{x}_j - \tilde{\boldsymbol{x}}\|$$ **Intf Cond**

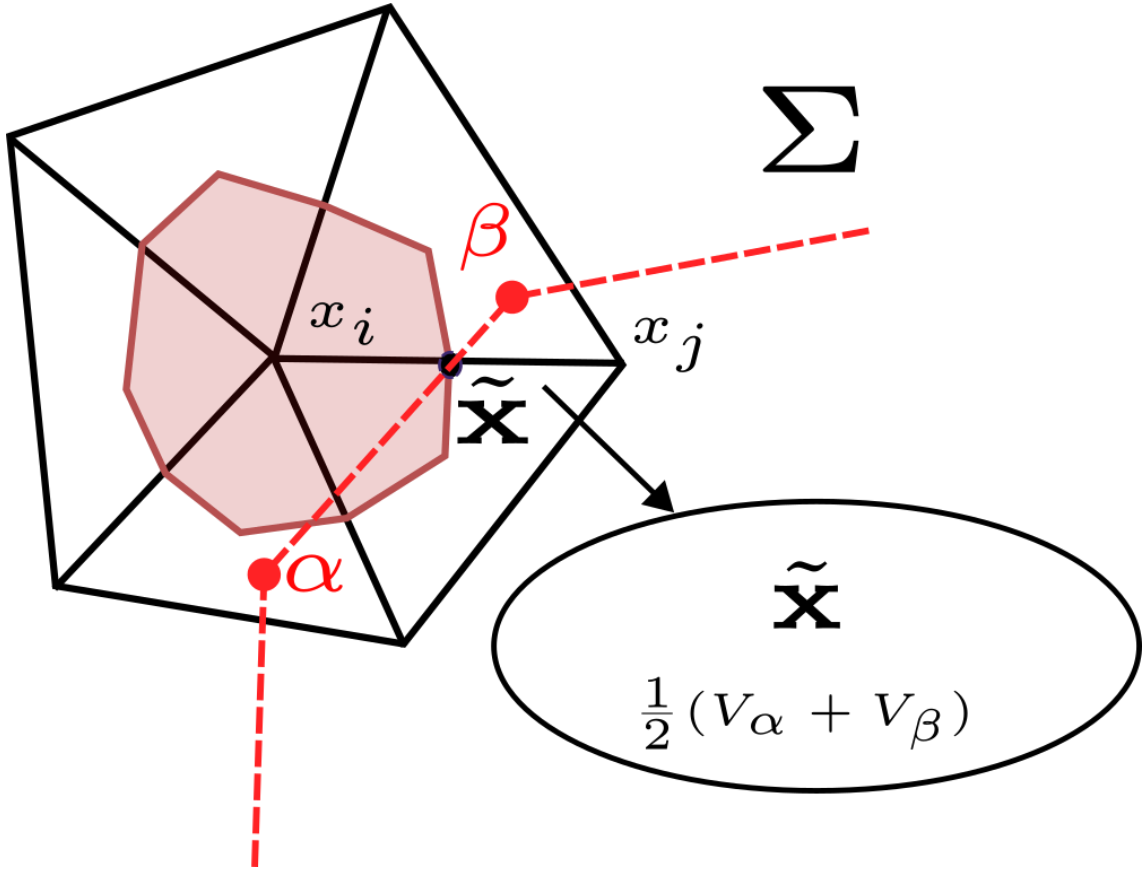
$$= V_m(\tilde{\boldsymbol{x}}) - \frac{[\![\sigma]\!]}{\sigma_{ex}}\left(\frac{\phi_j - [\![\phi]\!](\boldsymbol{x}_i) - \phi i}{\|\boldsymbol{x}_j - \boldsymbol{x}_i\|}\right)\|\boldsymbol{x}_j - \tilde{\boldsymbol{x}}\|$$



$$[\![\phi]\!](\boldsymbol{x}_j) = \frac{1}{1+\chi}V_m(\tilde{\boldsymbol{x}}) + \frac{\chi}{1+\chi}(\phi_j - \phi_i)$$

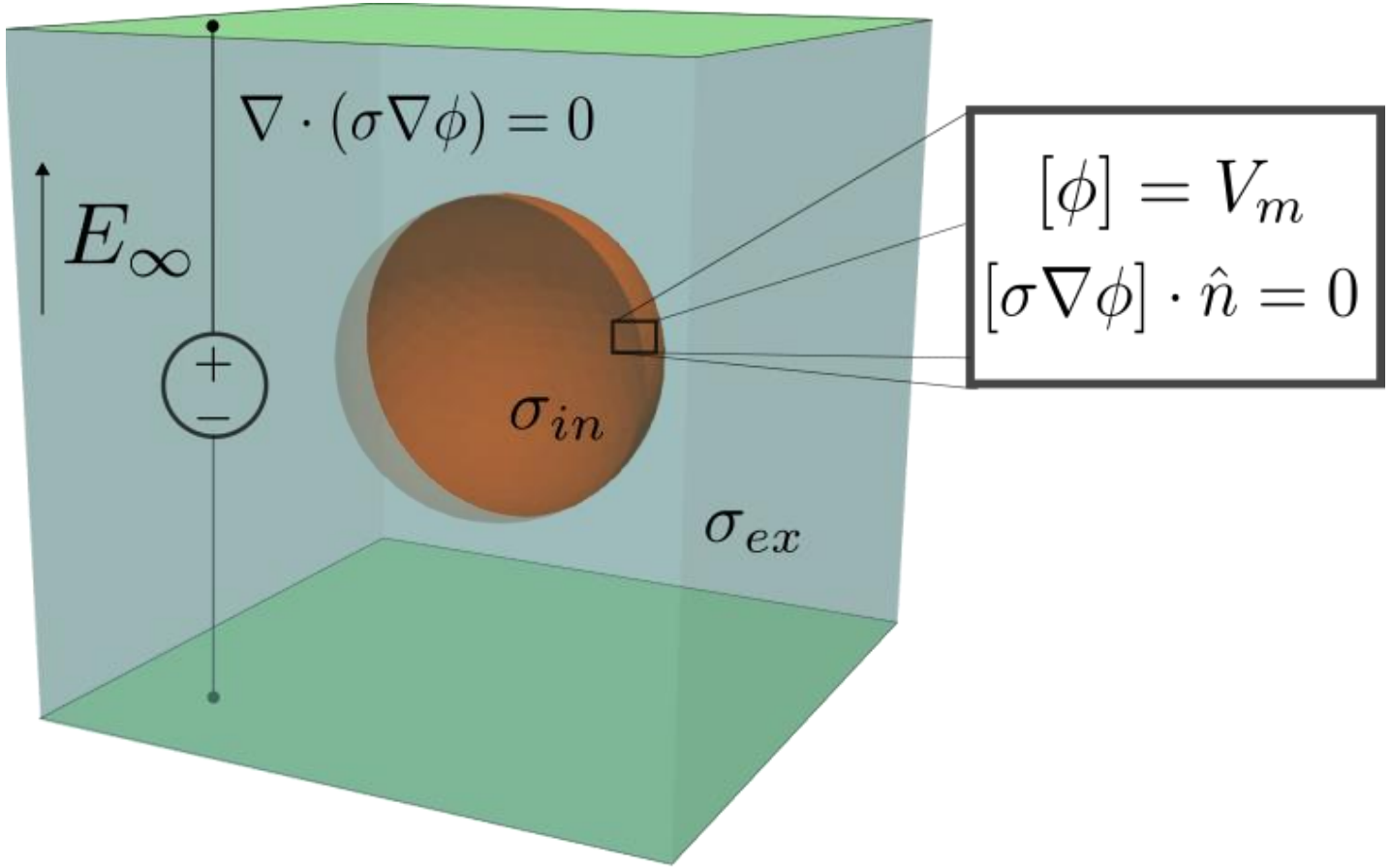
$$\chi = \frac{[\![\sigma]\!]}{\sigma_{ex}}\frac{\|\boldsymbol{x}_j - \tilde{\boldsymbol{x}}\|}{\|\boldsymbol{x}_j - \boldsymbol{x}_i\|}$$

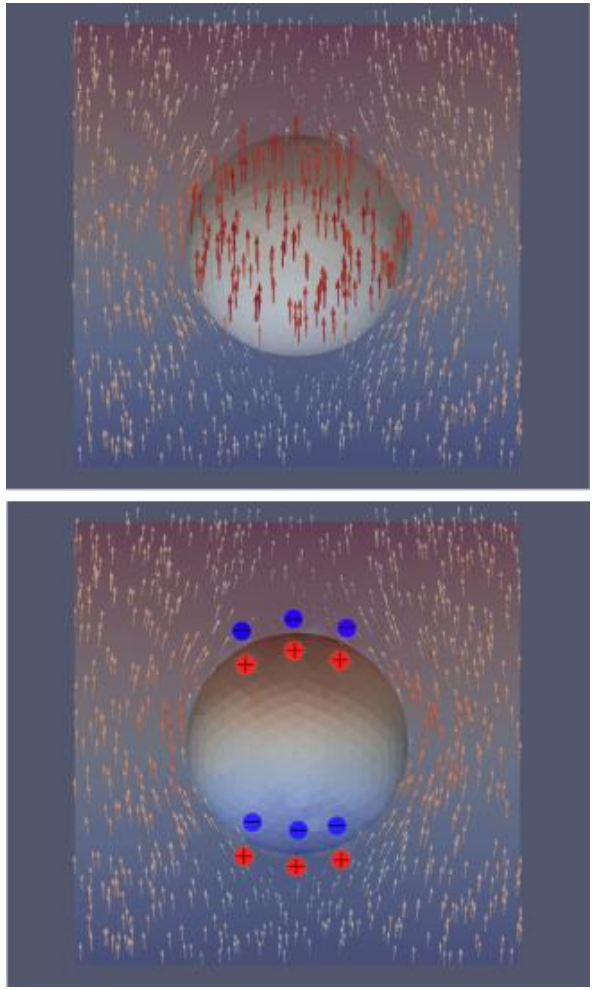
About Implementation



Loop on Cut Edges: Stock Intersection Position and Surface Jump

Prototype Case

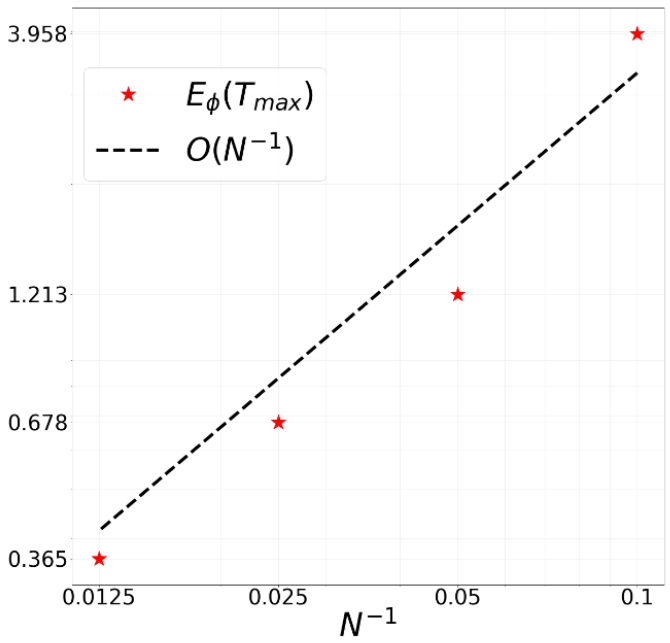


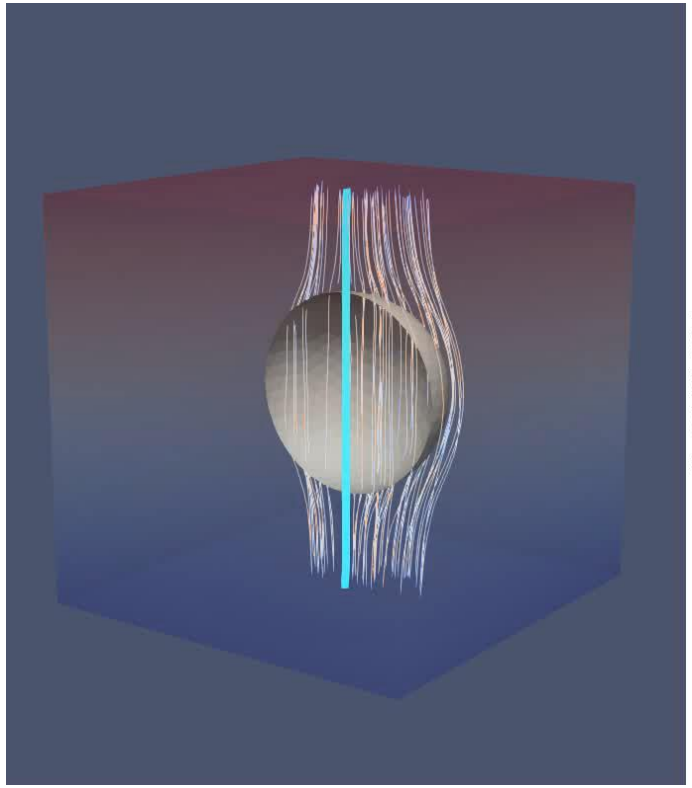


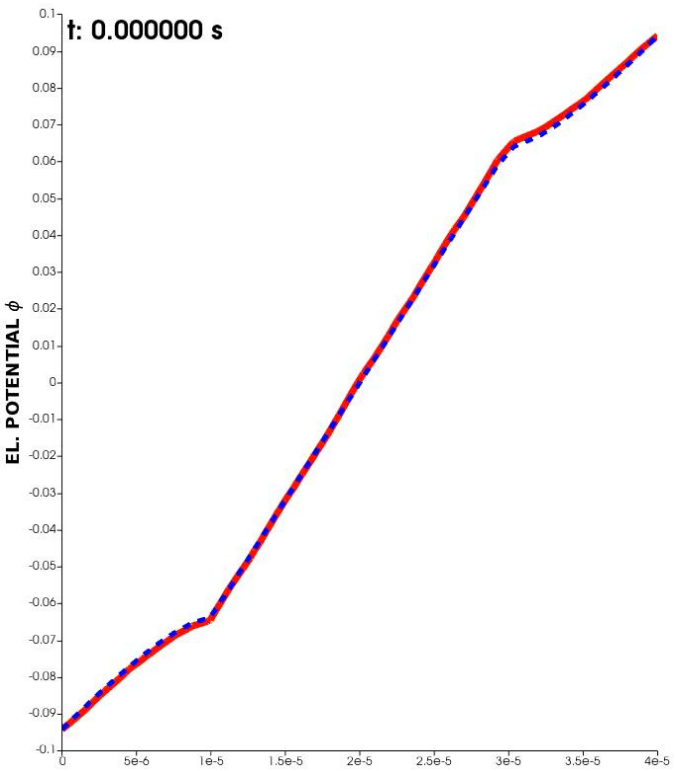
$$C_m \frac{\partial V_m}{\partial t} = \nabla \phi^{ex} \cdot \hat{n}$$

$$V_m|_{t=0} = 0$$

Prototype Case



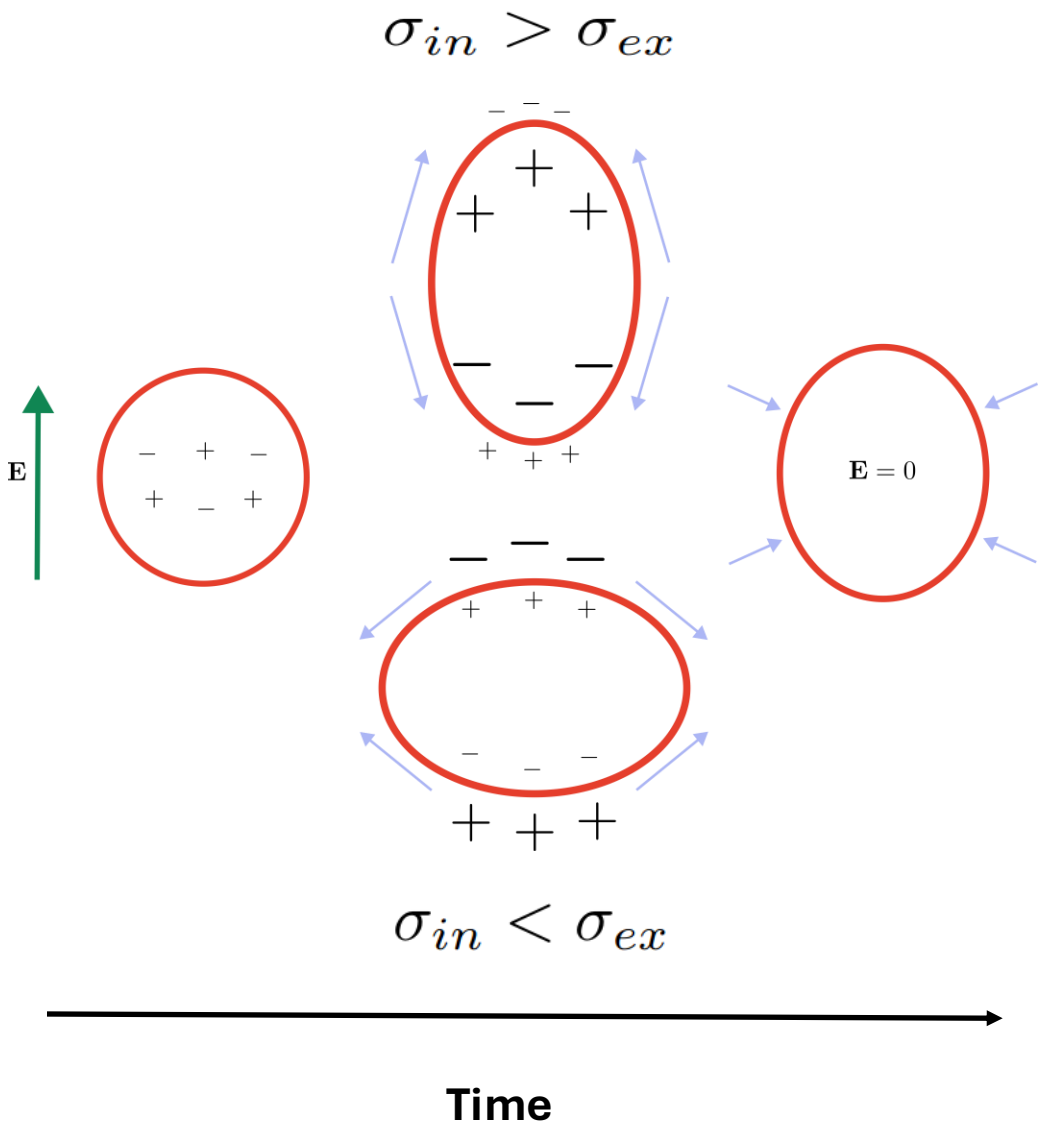


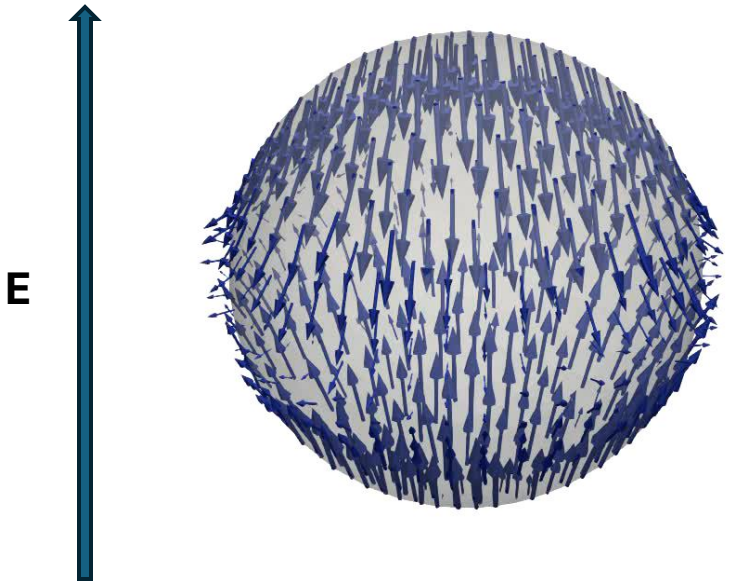


Sharp Jump Capture

ESTIMATED
REFERENCE

Deforming Membrane

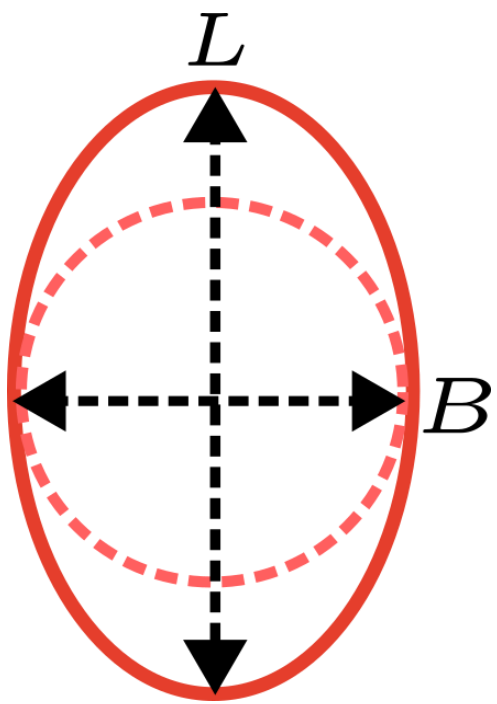


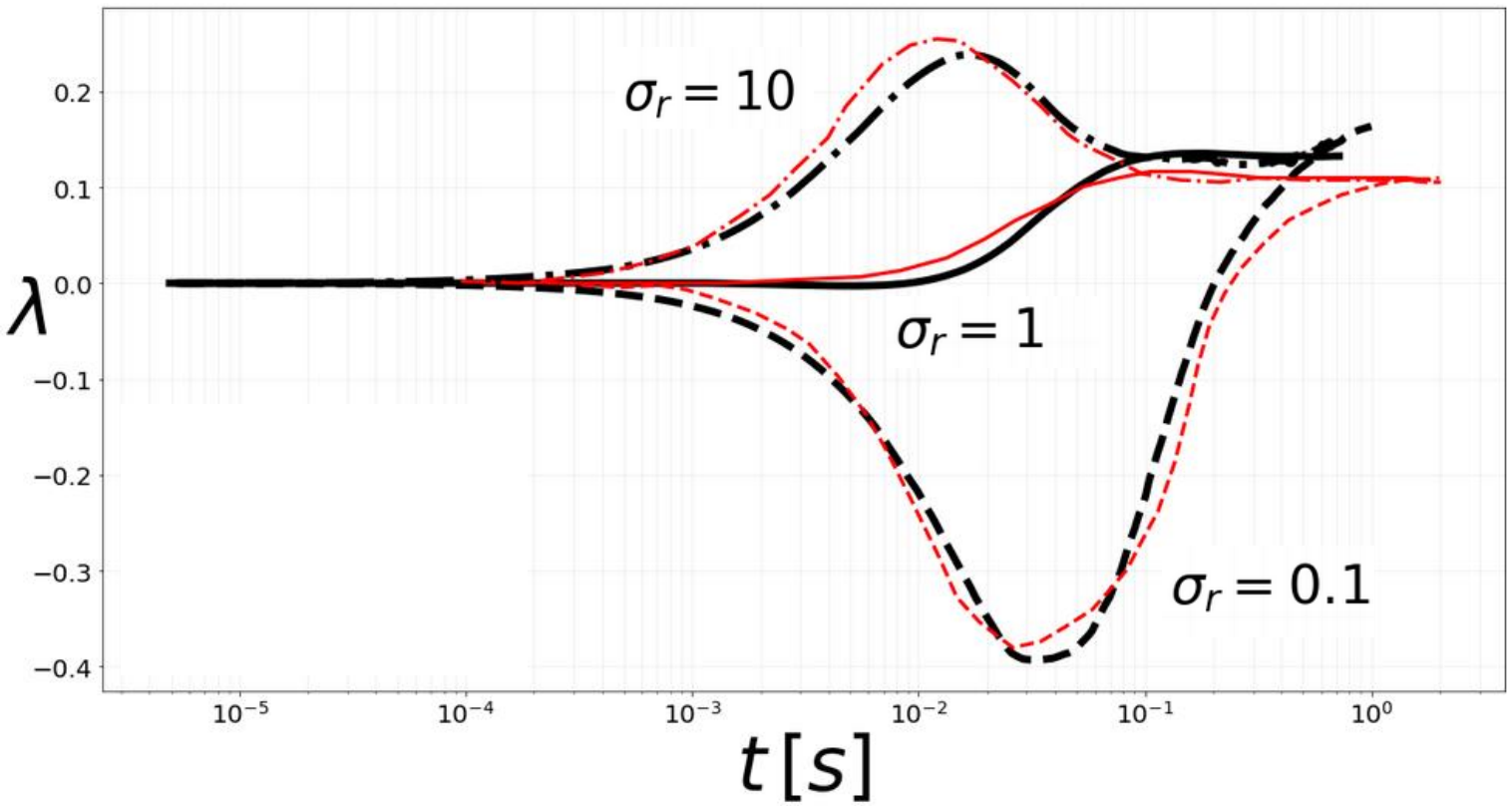


POP transition

Deforming Membrane

$$\lambda = \frac{L - B}{L + B}$$





Large deformation electrohydrodynamics of an elastic capsule in DC electric field

Sudip Das and Rochish M. Thaokar

Conclusion

GFM allows a representation of interface conditions compatible with the IBM engine

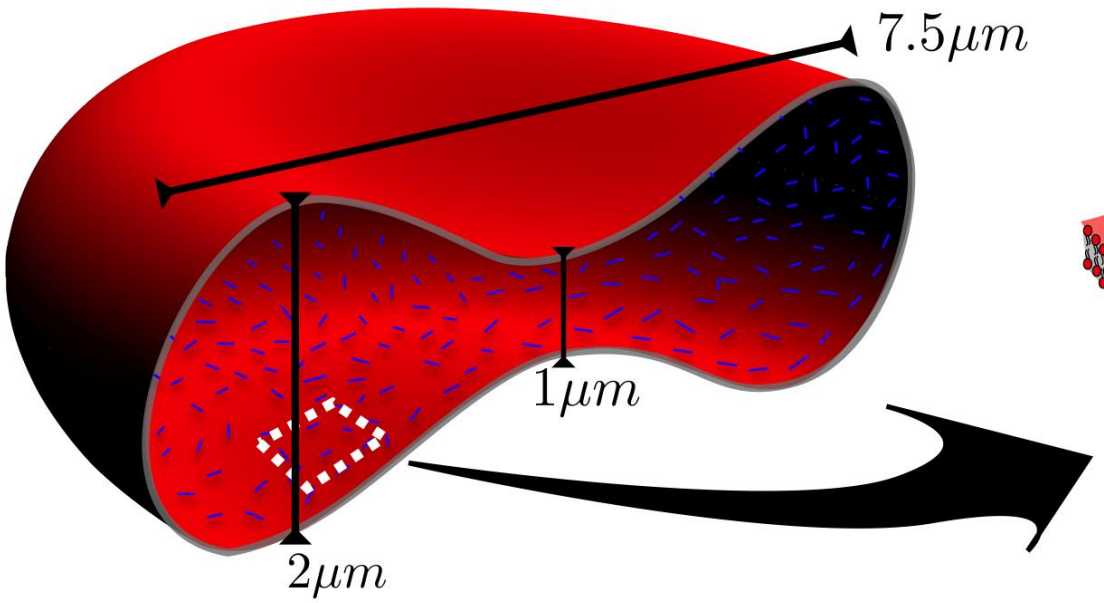
We are able to reproduce the charging of a fixed capacitive cell and the transition of a Skalak/Helfrich capsule undergoing Maxwell stress

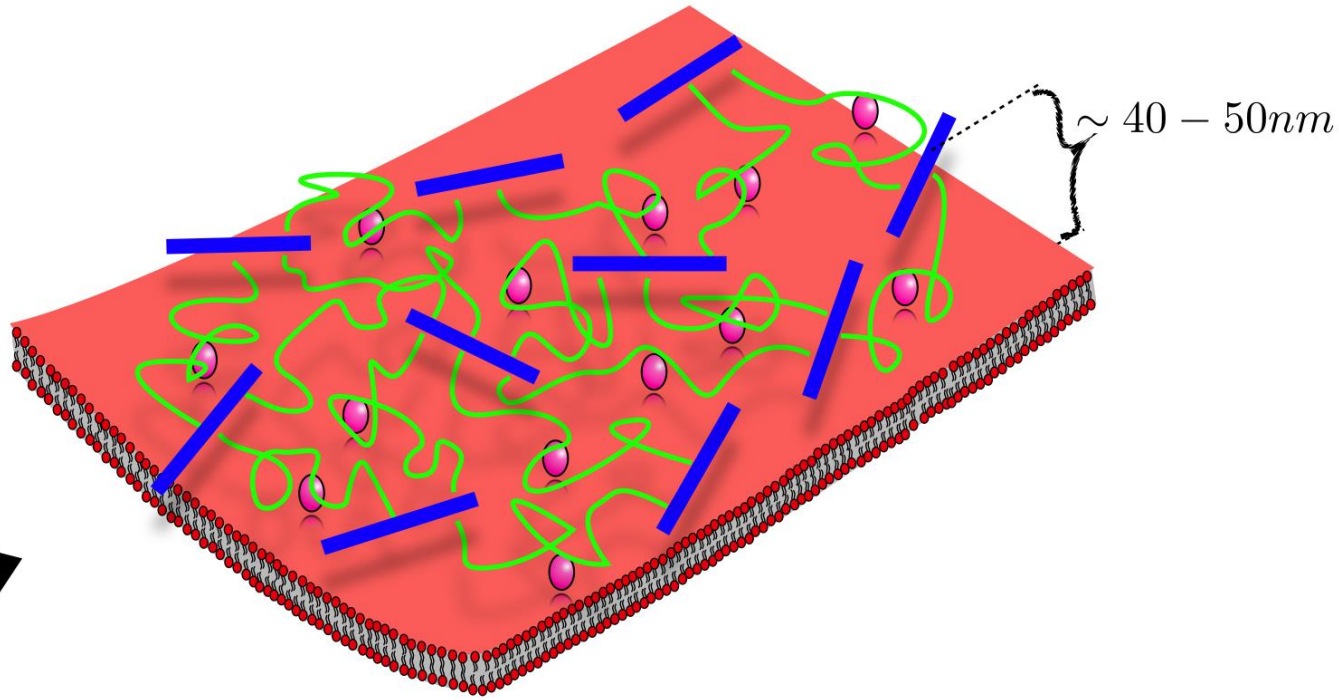
Thank you

Bonus Content

Red Blood Cells

Membrane Structure



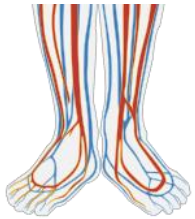


Junctional Complex 4.1

Ankyrin Complex

Spectrin Filament

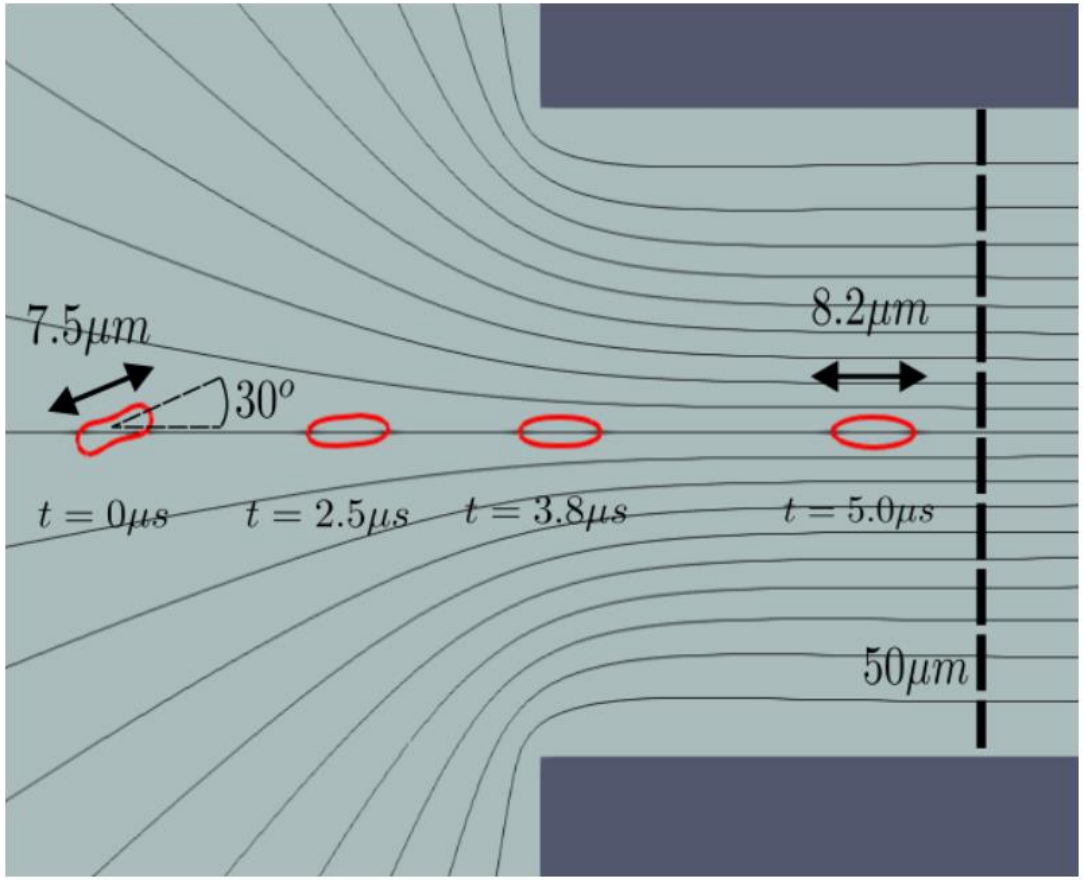
Lipid Bilayer



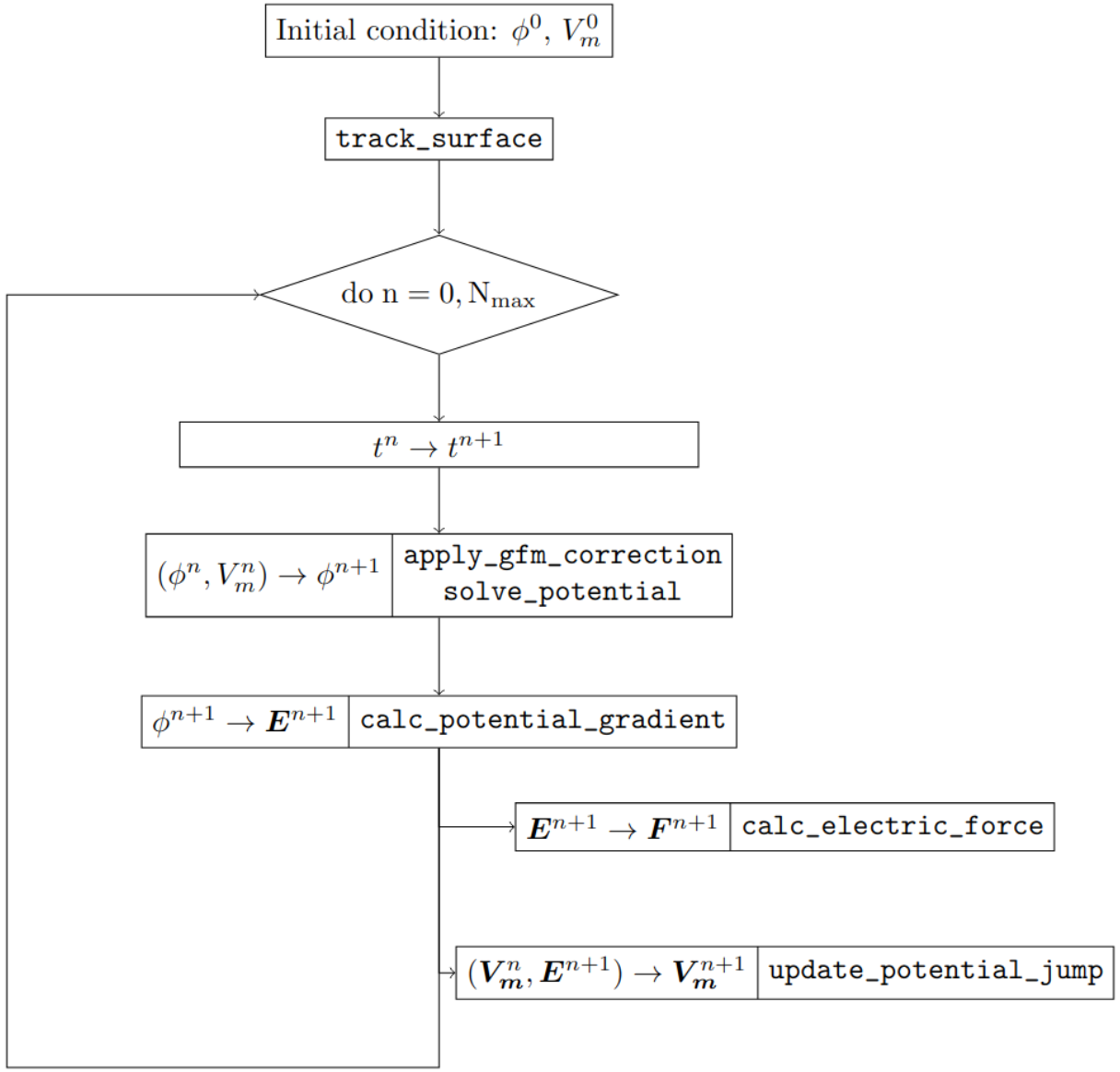
Cytoskeleton

Phospholipid

Towards RBCs

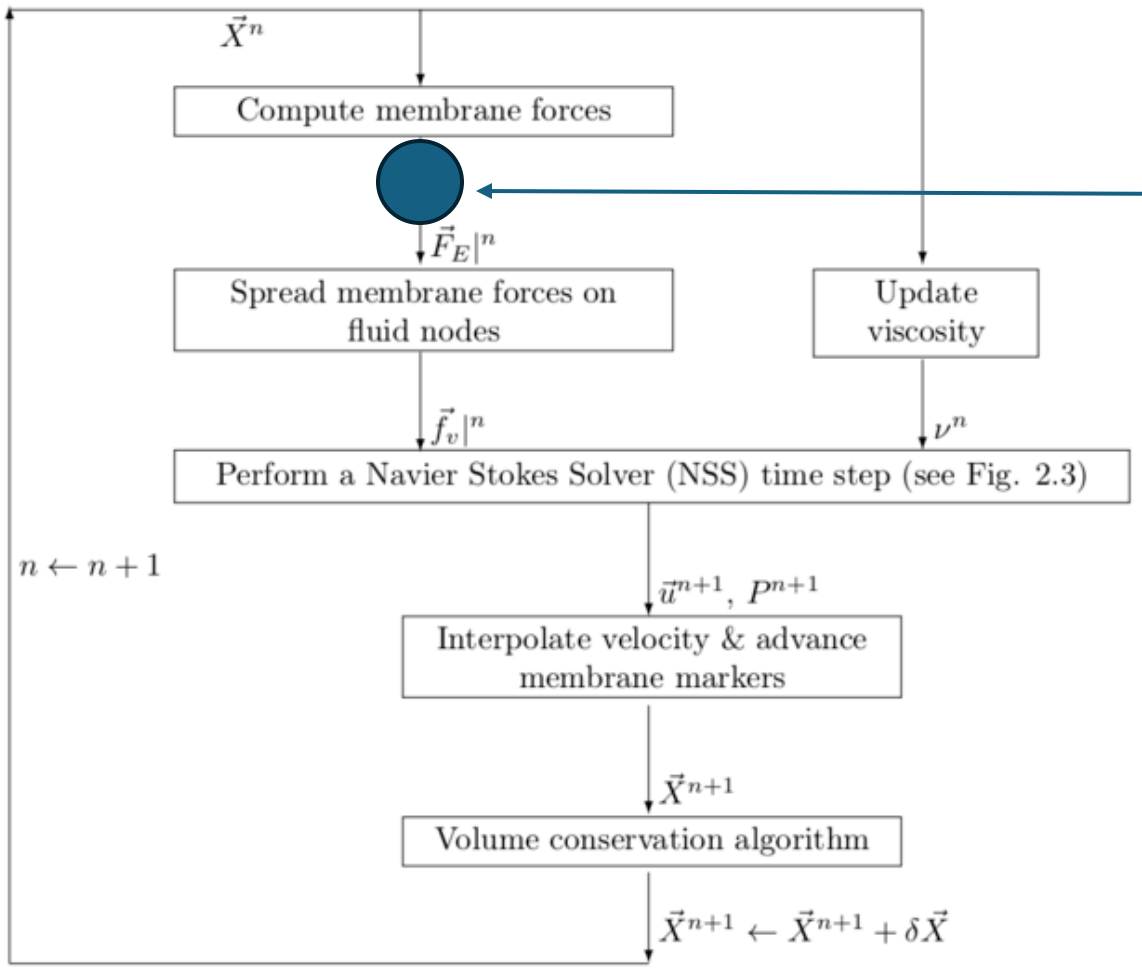


Overview


Initial condition: ϕ^0, V_m^0
track_surface
do n = 0, N$_{\text{max}}$
$t^n \rightarrow t^{n+1}$
$(\phi^n, V_m^n) \rightarrow \phi^{n+1}$
apply_gfm_correction
solve_potential
$\phi^{n+1} \rightarrow \boldsymbol{E}^{n+1}$
calc_potential_gradient
$\boldsymbol{E}^{n+1} \rightarrow \boldsymbol{F}^{n+1}$
calc_electric_force
$(\boldsymbol{V_m^n}, \boldsymbol{E}^{n+1}) \rightarrow \boldsymbol{V_m^{n+1}}$
update_potential_jump

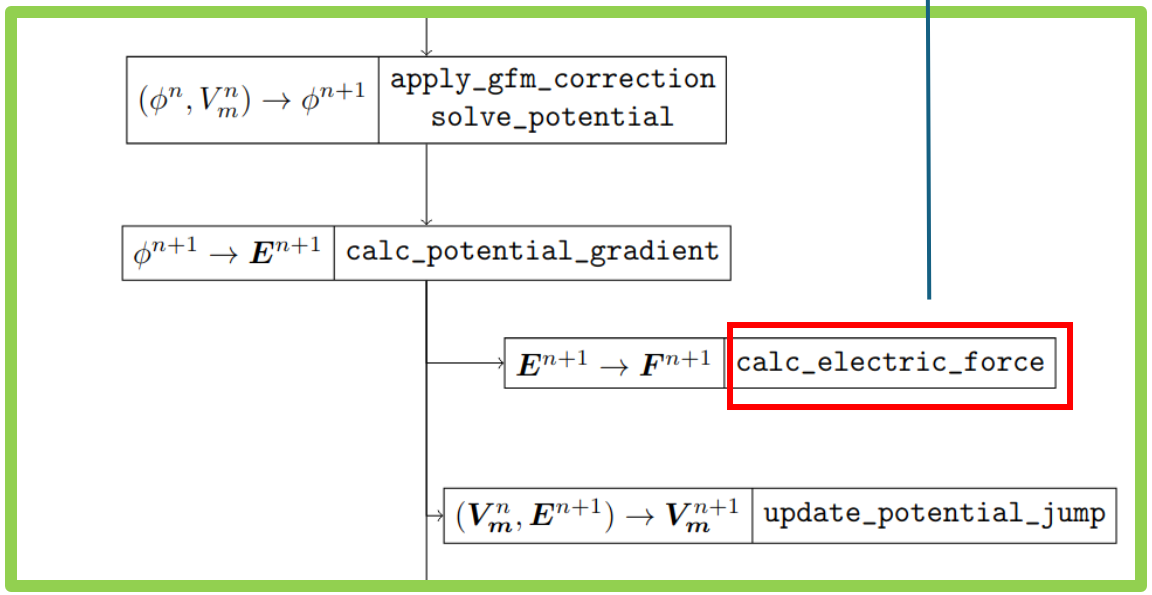
The Complete Picture

NS + Membrane Pipeline (RBC Solver)

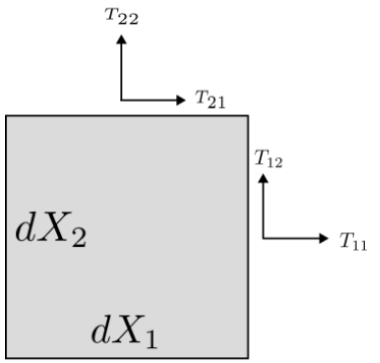


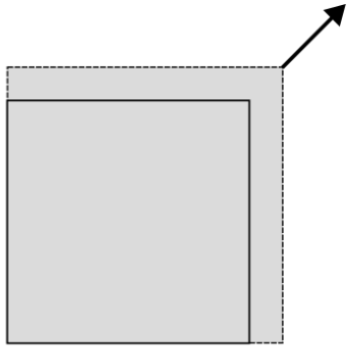
$$\boldsymbol{F} = \boldsymbol{F}_{memb} + \boldsymbol{F}_{maxwell}$$

Electrostatic Solver

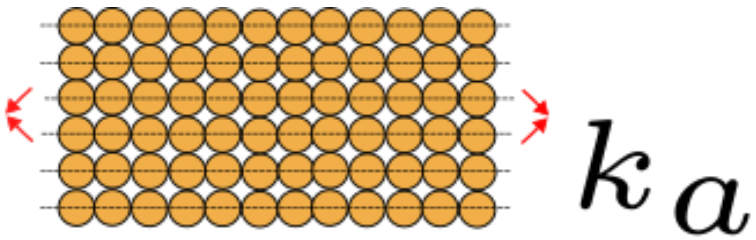


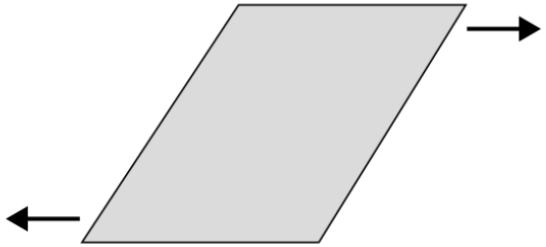
Skalak Energy



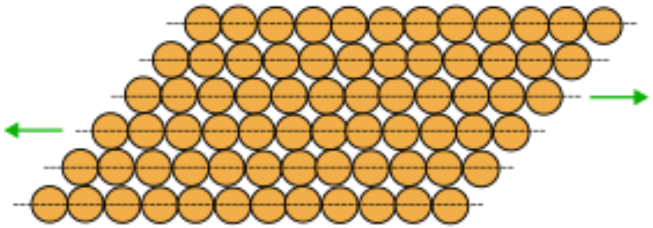


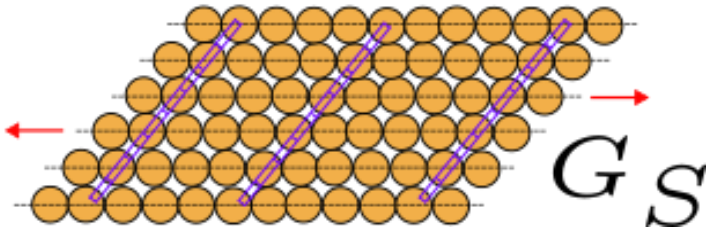
$dA = (1+\alpha)A_0$
Isotropic Stretching





$dA = 0$
Pure Shear





$$W_{SK} = \frac{G_{SK}}{4}[(\lambda_1^2+\lambda_2^2-2)^2 + 2(\lambda_1^2+\lambda_2^2-\lambda_1\lambda_2-1) + C(\lambda_1^2\lambda_2^2-1)^2]$$

Helfrich Energy

Picture resistance to bending

Bending Modulus

$$W_{hel} = \frac{k_b}{2} \int_{\Sigma} (H - C_0)^2 d\Sigma$$

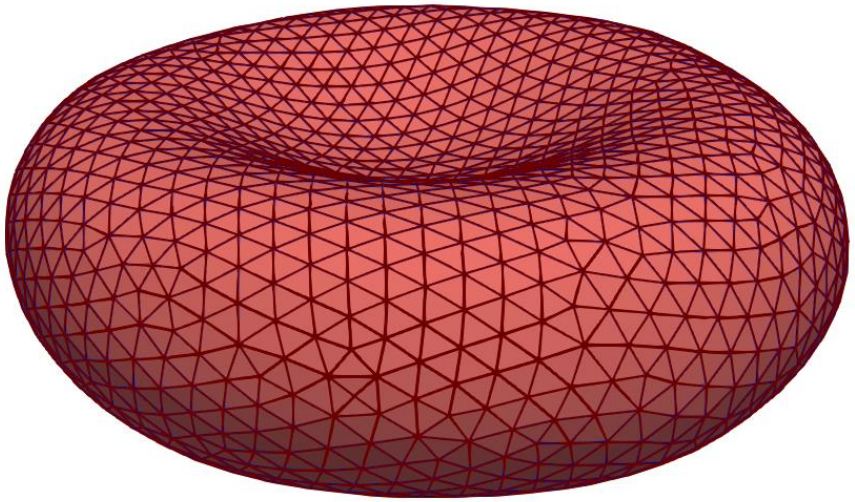
Mean Curvature

Spontaneous Curvature

Normal Bending Stress

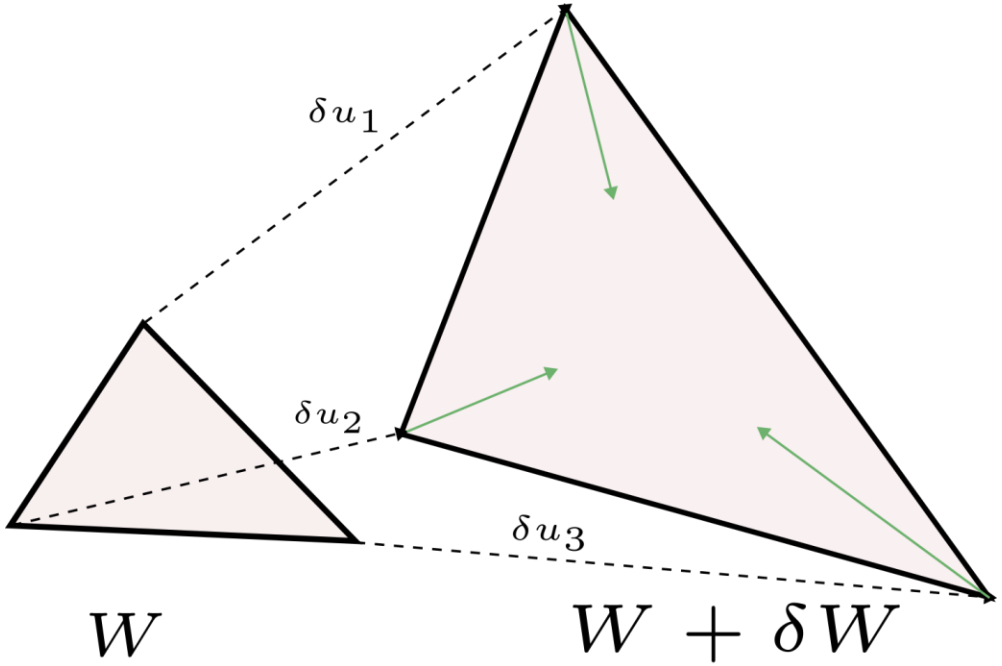
$$\boldsymbol{f}_b = k_b[(2H - C_0)(2H^2 - 2K + C_0 H) + 2\Delta_{LB} H]\boldsymbol{n}$$

Membrane Discretization



A mesh for the Surface

Principle of Virtual Works



$$\delta W = F_i \delta u_i$$

Forces live at nodes

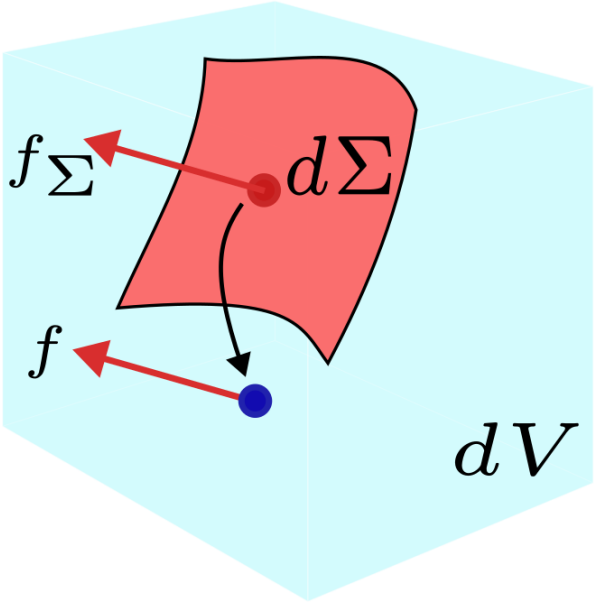
Fluid-Structure Interaction

NS in domain Ω

$$\begin{cases} \rho \frac{D\boldsymbol{v}}{Dt} + \nabla p + \mu \Delta \boldsymbol{v} = \boxed{\boldsymbol{f}} \\ \nabla \cdot \boldsymbol{v} = 0 \end{cases}$$

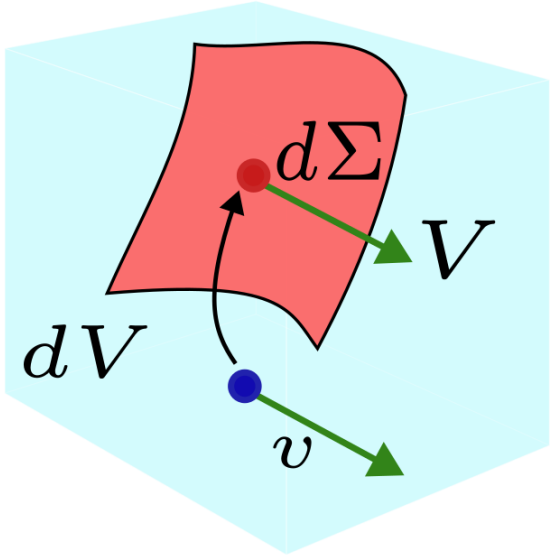
Force Transmission

$$\boldsymbol{f} = \int_{\Sigma} \delta(\boldsymbol{x} - \boldsymbol{X}) \boldsymbol{f}_{\Sigma}(\boldsymbol{X}) d\Sigma.$$



Velocity Interpolation (no-slip)

$$\boldsymbol{V}(\boldsymbol{X}) = \int_{\Omega} \delta(\boldsymbol{X} - \boldsymbol{x}) \boldsymbol{v}(\boldsymbol{x}) dV.$$

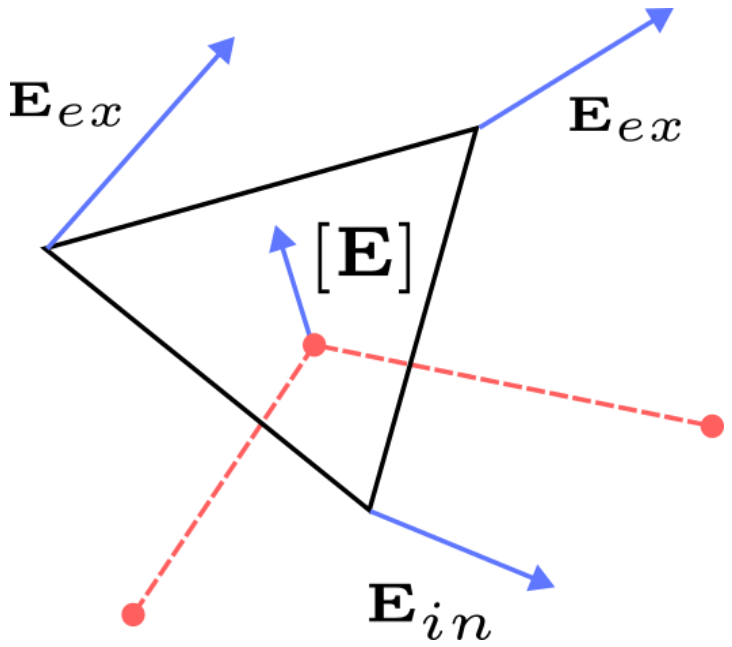


Jump Interpolation

$$C_m \frac{\partial V_m}{\partial t} + G_m V_m = \sigma_{in} \boldsymbol{E}_{in} \cdot \boldsymbol{n} = \sigma_{ex} \boldsymbol{E}_{ex} \cdot \boldsymbol{n}$$

Surface Variable **Volume Variable**

$$\boldsymbol{f}_{MW} = [\![\bar{\bar{T}}_{MW}]\!] \cdot \boldsymbol{n}$$

$$\bar{\bar{T}}_{MW} = \epsilon(\boldsymbol{E}\boldsymbol{E} - \frac{1}{2}\boldsymbol{E} \cdot \boldsymbol{E}\bar{\bar{I}})$$

Node-in-Element Interpolation